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ORIGINAL PAPER

Intelligent Multi-objective Regulation of Semi-active Joint Suspensions Equipped with Energy Harvesting Mechanisms

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Abstract

Background Semi-active vehicle suspension systems with integrated energy-harvesting mechanisms offer the potential to improve ride comfort and road holding while recovering energy. However, achieving these objectives simultaneously is challenging due to nonlinear dynamics, hybrid operating modes, and mixed constraints. Many existing approaches treat comfort, stability, and energy harvesting separately or rely on simplified linear models that do not fully capture the behaviour of energy-harvesting suspensions.

Methods A neural-network-based multi- objective control framework is proposed for a joint-internal semi-active vehicle suspension systemequipped with an energy harvesting mechanism. A high-idelity multibody model is irst developed in ADAMS to simulate the nonlinearsuspension dynamics and generate training data. A data-driven NARX neural network is then identiied to approximate the nonlinear dynamicsand replace the computationally intensive physical model in closed-loop simulations. The trained neural model is embedded in a neural-network-based model predictive control (NN–MPC) scheme, where sprung-mass acceleration, tire force, and harvested energy are explicitlyshaped in the cost function. Hybrid constraints associated with torque–velocity limits and harvesting-mode transitions are incorporated directlyinto the predictive optimization problem. The controller is evaluated via simulations under standardized stochastic road excitations.

Results Simulation results demonstrate that the proposed control framework effectively mediates the trade-off between ride comfort, road holding, and energy recovery. Compared with a passive suspension, the comfort-oriented coniguration reduces the RMS sprung-mass acceleration byapproximately 40–60% while maintaining tire-force levels within acceptable bounds. In the energy-harvesting mode, the recovered energyincreases signiicantly but is accompanied by higher suspension activity: RMS body acceleration rises from about 0.11–0.38 m/s² (passive) to0.17–0.60 m/s² (semi-active), and tire force increases from roughly 190–250 N to 330–810 N. Neural network validation shows fast errorconvergence and strong generalization performance, with regression coeficients exceeding 0.95, supporting the reliability of the identiiedmodel for predictive control.

Conclusions The proposed NN–MPC strategy provides a robust and scalable solution for energy-aware semi-active suspension systems. By integrating a data-driven NARX neural model with MPC and explicitly handling nonlinearities, mode switching, andmixed constraints, the controller effectively balances ride comfort, road holding, and energy harvesting. The results indicate strong potential forpractical implementation in electric and hybrid vehicles operating under diverse and uncertain road conditions.

Keywords Semi-active suspension system · Neural network · Model predictive control · Energy efficiency · Vehicle dynamic performance · Intelligent multi-objective

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Introduction

Rapid advancements in automotive technology and the increasing emphasis on energy-efficient mobility have transformed suspension systems into multifunctional subsystems that not only ensure ride comfort and road holding but may also contribute to energy recovery [1, 2]. Conventional passive suspensions—despite their inherent simplicity and reliability—cannot adapt to varying road excitations, limiting both comfort performance and dynamic stability [3, 4]. This limitation becomes particularly critical in electric and hybrid vehicles, where energy management directly affects driving range and overall efficiency [5]. As environmental concerns, fuel economy regulations, and global energy crises intensify, the demand for adaptive, intelligent suspension architectures continues to grow [6, 7].

Semi-active suspensions have emerged as a promising trade-off solution, offering many of the benefits of active systems with substantially lower energy consumption [8, 9]. They modulate damping properties in real time and maintain system passivity, enabling better handling and comfort without requiring high-power actuators [10, 11]. However, despite these advantages, semi-active systems still face challenges when multiple objectives—such as comfort, stability, and energy harvesting—must be optimized simultaneously under uncertain, nonlinear, and time-varying road conditions [12, 13].

Energy harvesting within suspension systems has become an active research direction aimed at capturing vibration energy otherwise wasted as heat in dampers [14, 15]. Various studies demonstrated that mechanical, electromagnetic, and hydraulic harvesters can provide tens to hundreds of watts depending on road roughness and vehicle speed [16, 17]. Nonetheless, simultaneous achievement of high-quality ride comfort and significant harvested power remains a difficult challenge due to inherent trade-offs between damping requirements for comfort and resistance levels needed for harvesting [18, 19].

Previous literature has investigated electromagnetic regenerative dampers [20], hydraulic regeneration circuits [21], and hybrid systems combining semi-active control with harvesting modules [22, 23]. Although these studies advanced the state-of-the-art, most of them either focused on single-objective optimization, simplified linear models, or relied on fixed-gain controllers unable to manage conflicting performance demands under real road scenarios [24, 25]. The lack of a unified multi-objective control strategy that simultaneously balances comfort, handling, and energy capture remains a major gap.

Furthermore, nonlinearities in suspension dynamics—such as geometric constraints, hysteresis, and time-varying damping—limit the accuracy of classical linear control approaches [26, 27]. Advanced model-based controllers, including sliding mode and H∞ designs, have provided robustness but often require precise real-time models and exhibit chattering or excessive computational cost [28, 29]. Moreover, multi-objective formulations with discrete constraints (such as damper switching or energy harvesting modes) lead to mixed-integer control structures that classical controllers cannot handle effectively [30, 31].

Neural-network-based controllers and learning-enabled architectures have recently emerged as powerful tools for capturing nonlinear behaviors and adapting to variable conditions in real time [32, 33]. Their ability to approximate unknown dynamics and adjust control policies based on experience makes them suitable for suspension systems involving nonlinear, multi-objective trade-offs [34, 35]. Yet, most neural-network controllers in existing literature are trained offline, lack generalization under high-frequency road disturbances, or do not integrate regenerative energy considerations into the training framework [36, 37].

A key limitation in previous research is the insufficient integration of joint-internal semi-active suspension configurations—systems where damping and energy harvesting elements are embedded within the suspension linkage rather than in traditional damper positions [38]. This novel architecture, referenced in the thesis file, enables more direct force transmission, improved packaging, and enhanced regenerative efficiency, but introduces new complexities such as rotational-link dynamics and hybrid switching constraints that have remained underexplored in related works [39].

Additionally, the combination of rotational control-link dynamics with regenerative harvesting circuits introduces mixed continuous–discrete control challenges. Maintaining stability and performance across harvesting modes (comfort–road-holding–energy maximization) requires advanced control allocation mechanisms [40, 41]. Previous work has rarely addressed this level of integrated control, particularly in a unified architecture that allows dynamic switching based on real-time performance indices [42].

The multi-objective nature of the problem—where improvement in one metric often leads to degradation in others—requires controllers capable of managing trade-offs adaptively. Traditional gain-scheduled or rule-based controllers cannot sufficiently navigate these conflicts, particularly when energy harvesting introduces nonlinear load coupling into the system [43, 44]. As a result, there is a need for a scalable control strategy that can dynamically shift priorities without compromising stability.

The work proposed in this research addresses the aforementioned gaps by developing a neural-network-based multi-objective controller tailored for a joint-internal semi-active suspension equipped with an energy harvesting module. The controller is trained to operate across multiple modes—comfort, road holding, and harvesting—while dynamically adjusting the system behavior based on real-time performance demands. This represents a significant departure from earlier approaches that treated these objectives independently.

A nonlinear dynamic model of the joint-internal suspension is employed, integrating rotational-link kinematics, hybrid damping behavior, and energy conversion characteristics. This modeling framework more accurately reflects real operation than conventional quarter-car models used in most prior studies [45, 46]. It also enables explicit representation of the torque–velocity constraints and switching surfaces governing the energy harvesting mechanism.

By using an adaptive neural architecture, the proposed controller not only captures nonlinearities but also handles mode-switching constraints inherently, without the need for manually defined threshold logic. This improves robustness against uncertain road excitations and enhances adaptability across scenarios, giving it a significant methodological advantage over classical and hybrid controllers reported in the literature [47, 48].

Most existing advanced suspension control strategies, including H∞, sliding-mode control (SMC), and adaptive model predictive control (MPC), fundamentally rely on explicit analytical state-space representations with continuously differentiable dynamics and smooth or convex control constraints. While such methods have demonstrated robustness against parametric uncertainties in conventional suspension layouts, their applicability becomes limited when rotational-link kinematics, hybrid damping–harvesting modes, and torque–velocity inequalities are present. In joint-internal semi-active suspension architectures, energy regeneration introduces mixed logical constraints and mode-dependent dynamics that lead to a hybrid nonlinear system structure, which cannot be accurately captured by a single smooth model. As a result, H∞ control typically requires multi-point linearization, losing robustness during mode switching, whereas SMC designs are prone to chattering under frequent torque sign changes and rotational inertia effects. Similarly, adaptive MPC formulations demand online model adaptation, which becomes ill-posed in the presence of discrete harvesting constraints and mixed-integer feasibility regions. These inherent limitations have been widely acknowledged in recent suspension and hybrid control studies, motivating the adoption of data-driven and learning-enabled control frameworks capable of handling nonlinear, hybrid, and multi-objective dynamics in an integrated manner [49–51].

Existing studies on regenerative and semi-active vehicle suspension systems have mainly focused on either model-based control strategies such as linear or nonlinear MPC, or data-driven neural-network-based approaches optimized for specific performance objectives. In MPC-based formulations, the energy harvesting mechanism is commonly represented through simplified convex force constraints or averaged efficiency models, which overlook the inherently hybrid and mode-dependent nature of torque–velocity relationships and dissipativity conditions in regenerative dampers [18, 23]. Consequently, these approaches often rely on heuristic switching rules or relaxed constraints that reduce physical fidelity during transitions between damping and energy harvesting modes. On the other hand, neural-network-based suspension controllers reported in the literature are typically trained offline for fixed operating conditions, lack explicit constraint handling, and do not explicitly address the mixed continuous–discrete feasibility regions induced by regenerative suspension dynamics [1, 25, 45].

Moreover, most existing studies adopt conventional damper-based or quarter-car suspension architectures [6, 14, 16], while recent reviews highlight the lack of control-oriented studies addressing more complex suspension layouts such as inertial or joint-internal configurations, where damping and energy harvesting are embedded within mechanical linkages and introduce additional nonlinear and hybrid behaviors [40, 43, 44]. As emphasized in recent review papers, there remains a clear research gap in integrating physically consistent energy regeneration models, hybrid constraint handling, and multi-objective control within a unified framework suitable for advanced semi-active suspensions [40, 47].

To address these gaps, this paper proposes a unified neural-network-assisted mixed-integer MPC framework specifically developed for a joint-internal semi-active suspension with integrated energy harvesting. Unlike existing MPC-based approaches, the proposed formulation explicitly models torque–velocity switching surfaces and dissipativity constraints as mixed-integer feasibility regions, ensuring physically consistent operation during energy harvesting mode transitions. Unlike purely data-driven methods, the neural network is embedded as a structured nonlinear predictor within the MPC loop rather than acting as a black-box controller, thereby preserving constraint awareness and optimization transparency. Furthermore, the proposed controller simultaneously balances ride comfort, road holding, and energy recovery, and the trade-offs among these conflicting objectives are quantitatively evaluated using ISO-normalized performance indices under realistic road excitations. By combining joint-internal suspension mechanics, mixed-integer constraint handling, and learning-enhanced predictive control in a single coherent framework, this study advances beyond existing MPC- or NN-based regenerative suspension controllers and contributes a scalable and physically consistent solution for next-generation energy-aware vehicle suspension systems.

Dynamic Modeling of Vehicle Suspension System

A modern vehicle suspension must ensure ride comfort, road-holding, and vibration isolation under various road profiles.

The most fundamental mechanical configuration used for theoretical modeling is the quarter-car model, representing one wheel assembly and its associated suspension components.

The dynamic model consists of two main masses:

- the sprung mass m_s (vehicle body portion supported by suspension), and.
- the unsprung mass m_u (wheel, hub, and part of the suspension arm).

These masses are connected through a spring-damper assembly with stiffness $k_{\&}$ and damping coefficient c_s. The wheel-road contact is modelled by another spring of stiffness k_q representing tire compliance.

The equations of motion describing vertical dynamics are:

$$m_s\ddot{z}_s + c_s\left(\dot{z}_s - \dot{z}_u\right) + k_s\left(z_s - z_u\right) = F_a \tag{1}$$

$$\begin{gathered} m_u\ddot{z}_u - c_s\left(\dot{z}_s - \dot{z}_u\right) - \\ k_s\left(z_s - z_u\right) + k_q\left(z_u - z_T\right) = -F_a \end{gathered} \tag{2}$$

where F_a is the actuator or semi-active control force, and z_τ is the road displacement excitation.

To capture rotational damping effects found in real jointed suspensions, we map linear damping c_s to its rotational equivalent c_{F}:

$$M_r = c_r\dot{\theta} = r^2 c_s\dot{\theta} \tag{3}$$

where r is the arm length connecting the rotational and linear damping domains.

1. **Extended Half-Car and Full-Car Model**

To investigate the system's pitch and bounce responses, a half-car model expands the single-wheel system into two suspensions (front and rear), with the chassis modeled as a rigid body with pitch inertia $\boldsymbol{I_y}$.

The vertical and angular motions are given by:

$$\begin{gathered} m\ddot{z}_s + c_f\left(\dot{z}_s - \dot{z}_{uf}\right) + k_f\left(z_s - z_{uf}\right) + \\ c_r\left(\dot{z}_s - \dot{z}_{ur}\right) + k_r\left(z_s - z_{ur}\right) = F_{af} + F_{ar} \end{gathered} \tag{4}$$

$$\begin{aligned} I_y\ddot{\theta} = {} & l_f\left[c_f\left(\dot{z}_s - \dot{z}_{uf}\right) + k_f\left(z_s - z_{uf}\right) - F_{af}\right] \\ & -l_r\left[c_r\left(\dot{z}_s - \dot{z}_{ur}\right) + k_r\left(z_s - z_{ur}\right) - F_{ar}\right] \end{aligned} \tag{5}$$

where:

- l_f and l_r are distances from the center of gravity to the front and rear axles, respectively;
- subscripts f and r denote front and rear assemblies.

This formulation enables simulation under asymmetric road excitations (front-rear phase delay), which are highly sensitive to vehicle pitch and energy dissipation mechanisms.

In this study, the use of multiple modeling fidelities follows a hierarchical and physics-preserving modeling strategy, rather than independent or re-parameterized formulations. Specifically, the quarter-car model is employed at the control-oriented level to derive analytical expressions for suspension force, relative motion, and harvested energy, where the dominant vertical dynamics and energy flow mechanisms can be isolated and clearly interpreted. This level enables tractable formulation of the MPC cost function and constraints without loss of physical meaning.

The half-car model extends the same suspension force and energy formulation to include pitch dynamics by introducing the longitudinal distribution of the sprung mass and suspension attachment points. Importantly, no suspension parameters are re-identified at this stage; instead, the quarter-car parameters (spring, damper, rotary actuator, and energy harvesting relations) are directly mapped to the front and rear suspension units using geometric scaling and moment equilibrium, thereby preserving energy consistency and force transmission characteristics.

Finally, the full-vehicle ADAMS model serves as a high-fidelity validation environment rather than a redesign of the suspension physics. The same suspension architecture, rotary damper characteristics, friction models, and torque–speed relationships are implemented at each wheel corner, while additional effects—such as joint friction, parasitic losses, lateral load transfer, and coupled longitudinal–lateral dynamics—are naturally introduced by the multibody formulation. As demonstrated in Fig. 6, the close agreement between the nonlinear analytical model and the ADAMS simulation confirms that the hierarchical transfer preserves the essential dynamics and dissipation mechanisms across modeling levels.

Therefore, the transition from quarter-car to half-car and ultimately to full-vehicle modeling should be interpreted as a progressive enrichment of system dynamics, rather than a change in suspension physics or parameter definitions. This hierarchical approach is widely adopted in vehicle dynamics and control studies to balance analytical transparency, controller design feasibility, and high-fidelity validation.

Active Control

In active systems, external energy sources generate control forces F_a to directly counteract disturbances. The ideal control goal is to minimize sprung-mass acceleration while respecting actuator limitations:

$$J_1 = \int_0^T \overrightarrow{z}_s^2 dt \; subject\ to\ |F_a| \leq F_{\max} \tag{6}$$

In a fully active suspension, an external power actuator (e.g., hydraulic, electro-hydraulic, or pneumatic) applies a bidirectional force F_a between the sprung and unsprung masses. The active force is computed using a feedback control law:

$$F_a = K_z (z_s - z_u) + K_i (\dot{z}_s - \dot{z}_u) + K_{ext} z_r \tag{7}$$

Where: $K_z, K_{\dot{z}}$, and K_{ext} are feedback gains determined by optimal control design (e.g., LQR or MPC). The first term compensates relative displacement, the second term damps relative velocity, and the last term anticipates road preview compensation if road estimation sensors are available.

The total damped power dissipated by the actuator is:

$$P_{act}(t) = F_a \cdot (\dot{z}_s - \dot{z}_u) \tag{8}$$

A positive P_{act} means the actuator injects energy into the system (active work), whereas a negative value indicates energy regeneration or damping action.

4. Semi-active Control

Semi-active suspensions vary their damping coefficient $c(t)$ in real time based on state feedback, without consuming large external energy.

The main control law for variable damping is defined through the groundhook, skyhook, or hybrid algorithms.

For instance, a skyhook-based semi-active damper controller is formulated as:

$$\begin{array}{l} c(t) = c_{\max}, \text{ if} (\dot{\text{z}}_s - \dot{z}_u)(\dot{\text{z}}_s) > 0 \\ \qquad c_{\min}, \text{ otherwise} \end{array} \tag{9}$$

This formulation ensures damping only when energy dissipation benefits vibration reduction.

In semi-active suspensions, no external energy source continuously injects power; instead, a variable damper changes its coefficient $c(t)$ in real-time to modulate forces passively.

The damping force F_d in the controllable damper is defined by:

$$F_d = c(t)(\dot{z}_s - \dot{z}_u) \tag{10}$$

with bound constraints:

$$c_{\min} \leq c(t) \leq c_{\max} \tag{11}$$

Depending on control logic, $c(t)$ is tuned according to relative displacement and velocity, aiming to minimize sprung mass acceleration while ensuring energy balance.

To guarantee that a semi-active damper never injects energy, its damping force is constrained so that:

$$F_d (\dot{z}_s - \dot{z}_u) \geq 0 \tag{12}$$

This ensures that the damper always absorbs mechanical energy instead of producing motion.

A typical piecewise control logic uses:

$$\begin{array}{l} F_d = \{C_{max} (\dot{z}_s - \dot{z}_u),\ if\ (\dot{z}_s - \dot{z}_u)(\dot{z}_s - \dot{z}_r) > 0] \\ \qquad\qquad C_{min} (\dot{z}_s - \dot{z}_u),\ otherwise \end{array} \tag{13}$$

Equation (13) corresponds to the skyhook control law, which approximates the ideal "virtual damper to the sky"-damping body motion rather than wheel motion.

For improved performance across both ride comfort and road holding, a blended controller combines skyhook and groundhook strategies:

$$F_d = \eta_s c_s (\dot{z}_s - \dot{z}_u) + \eta_g c_t (\dot{z}_u - \dot{z}_r) \tag{14}$$

Where: η_s and η_g are adaptive weighting coefficients depending on driving conditions; for example:

- $\eta_s \to 1, \eta_g \to 0 \to$ smooth road (focus on comfort),
- $\eta_s \to 0, \eta_g \to 1 \to$ rough road (focus on grip and tire contact).

The real damper (such as magnetorheological or electrorheological) converts commanded damping levels into physical resistance by controlling the fluid's viscosity via current $I(t)$:

$$c(t) = c_0 + \alpha I(t) \tag{15}$$

where α is the sensitivity factor ($\text{N} \cdot \text{s/m/A}$).

The net control force acting between the sprung and unsprung masses at any time is the summation:

$$F_{net} = F_s + F_d + F_a \tag{16}$$

with:

$$F_s = k_s\left(z_s - z_u\right),\ F_d = c\left(t\right)\left(\dot{z}_s - \dot{z}_u\right), \\ F_a = control\ actuator\ output$$

Depending on the control type:

- In passive systems, $F_a = 0, c\left(t\right) = c_s =$ const.
- In semi-active, $F_a = 0, c\left(t\right)$ variable within bounds.
- In active, both c_s and F_a are adaptive or fully generated by the controller.

The total control objective aims to minimize both vibration energy and actuator energy consumption:

$$J = \int_0^T \left[w_1 \dot{z}_s^2 + w_2\left(z_u - z_r\right)^2 + w_3 F_a^2\right] dt \tag{17}$$

where weighting factors w_1, w_2, w_3 are tuned based on desired trade-offs (comfort vs. road-holding vs. energy).

In the semi-active case with energy recovery, the regenerative power P_{rec} subtracted from F_a 's cost term modifies (17) into:

$$J^* = \int_0^T \\ \left[w_1 \dot{z}_s^2 + w_2\left(z_u - z_r\right)^2 + w_3\left(P_{act} - P_{rec}\right)\right] dt \tag{18}$$

This form connects directly to the control energy optimization used in the MPC-neural hybrid controller.

Nonlinear Road Excitation Modeling

To realistically simulate vibrational conditions in urban and off-road driving, the road disturbance is modeled as a stochastic process using band-limited white noise.

The equivalent road profile $z_r\left(t\right)$ is generated via first-order shaping filter:

$$\dot{z}_r\left(t\right) = -\alpha\, z_r\left(t\right) + w\left(t\right), \\ S_{zr}\left(\omega\right) = \frac{S_0}{1+\left(\frac{\omega}{\omega_0}\right)^2} \tag{19}$$

where: S_0 : road roughness coefficient (ISO road classification), ω_0 : cut-off angular frequency, $w\left(t\right)$: Gaussian white noise process.

Different values of S_0 represent various road grades (A-E), simulating higher vibration intensities at rougher surfaces.

The regenerative damper is designed to convert relative motion between suspension nodes into electrical energy using an electromagnetic transducer.

The power harvested at any instant is:

$$P\left(t\right) = \eta\, F_d \dot{z} = \eta\, c\left(t\right)\left(\dot{z}_s - \dot{z}_u\right)^2 \tag{20}$$

where η denotes the conversion efficiency of the electromechanical system.

The total energy recovered over simulation period T is:

$$E_{\text{rec}} = \int_0^T P\left(t\right) dt \tag{21}$$

Control of $c\left(t\right)$ thus serves a dual objective: improving ride comfort and maximizing energy recovery.

Model Identification Using Neural Network (NARX Model)

For accurate real-time control, the nonlinear suspension dynamics are approximated using a Neural Network Auto-Regressive model with eXogenous inputs (NARX).

The discrete model structure is given by:

$$y\left(k+1\right) = \\ f\left(y\left(k\right), y\left(k-1\right), \ldots, u\left(k\right), u\left(k-1\right), \ldots\right) + e\left(k\right) \tag{22}$$

where: $y\left(k\right)$ is the output (sprung mass displacement or acceleration), $u\left(k\right)$ is the control signal (actuator or damping command), $f(\cdot)$ is a nonlinear mapping realized by a trained neural network.

In the simulation, data from the ADAMS dynamic model are used as the training dataset. The neural network is trained using Levenberg-Marquardt optimization, with six hidden neurons ensuring fast convergence and accurate generalization.

After training, the NARX model replaces the physical ADAMS plant in MATLAB/Simulink for closed-loop control validation.

The neural NARX model realizes the nonlinear mapping $\phi\left(\cdot\right)$ through a neural network of structure:

$$\phi\left(x\right) = T\left(x\right) + f\left(Q\left(x\right)\right) + d \tag{23}$$

where: $T\left(x\right)$: Linear transformation (often a direct weighted connection from inputs to outputs), $f\left(Q\left(x\right)\right)$: Nonlinear component approximated by hidden neurons, d

: bias term, $Q(x)$: input vector containing delayed inputs and outputs.

The nonlinear transformation of the hidden layer is usually given by a sigmoid activation:

$$\begin{aligned} f(Q(x)) &= \sum_{k=1}^{m_k} a_k g\left(b_k^{\top} Q(x)\right) \\ g(s) &= \frac{1}{1+e^{-s}} \end{aligned} \tag{24}$$

with: n_h : number of hidden neurons, a_k, b_k : weights to output and input layers.

Thus, for training, the network has two layers: Hidden layer with nonlinear activations $g(s)$; Output layer with linear combination.

This structure allows the NARX network to approximate any smooth nonlinear mapping.

For Model Predictive Control (MPC) or analytical interpretation, the nonlinear NARX can be locally linearized around the operating point k :

$$\begin{aligned} y(k) = \sum{}_{l=1}^{n} b_l(k)\, u(k-l) - \\ \sum{}_{l=1}^{n_r} a_l(k)\, y(k-l) + d(k) \end{aligned} \tag{25}$$

and for one-step prediction:

$$\begin{aligned} \widehat{y}(k+1 \mid k) = \sum{}_{i=1}^{n_k} b_i(k)\, u(k-i+1) \\ -\sum{}_{j=1}^{n_k} a_j(k)\, y(k-j+1) + d(k) \end{aligned} \tag{26}$$

This is consistent with Eqs. (21−2) to (24−2) extracted from your file; coefficients $a_i(k), b_i(k)$ are adaptively estimated by the neural model.

In prediction horizon p, the recursive form yields:

$$\begin{aligned} \widehat{y}(k+p?k) = \mathbf{B}_p(k)\, \mathbf{U}_p(k) \\ -\mathbf{A}_p(k)\, \mathbf{Y}_p(k) + d(k) \end{aligned} \tag{27}$$

where $\mathbf{B}_p(k)$ and $\mathbf{A}_p(k)$ are the matrices of local linear approximations at step k.

The training minimizes the sum of squared errors (SSE) between measured and predicted outputs:

$$\mathrm{SSE} = \sum{}_{k\in \text{ dataset}} \left(y(k) - \widehat{y}(k?k-1)\right)^2 \tag{28}$$

Weights are updated via Levenberg-Marquardt or Adam algorithms:

$$W_{\text{new}} = W_{\text{old}} - \eta \left[J^T J + \lambda I\right]^{-1} J^T e \tag{29}$$

where: J: Jacobian of prediction errors w.r.t. weights, e : vector of residuals, λ: damping term, η: learning rate.

II. Neural Network Training and System Identification

Given the high precision of the ADAMS simulation model, it is feasible to employ this model within the control policy as a predictive framework. Yet, due to computational complexity, reliance on such a detailed analytical model is impractical for real-time control or extensive studies. A convenient alternative is to derive a simplified model of the system behavior through the use of a **neural network**. This approach makes it possible to capture the nonlinear response of the suspension efficiently, providing approximate predictions suitable for use in iterative control algorithms.

Neural Network Learning Process.

Although numerous software tools can be employed for constructing neural network models, the major advantage lies in their ability to represent extremely complex or nonlinear relationships between inputs and outputs with minimal data requirements and acceptable adaptability. In this section, the process of training the neural network and justification for employing it in dynamic system identification are provided.

Reasons for Employing Neural Networks in System Identification.

- **Nonlinear Relationship Modeling**

Neural networks can accurately replicate nonlinear and highly complex interactions between varying parameters. This property is essential when the analytic form of the dynamic system is either unknown or too involved to express mathematically. For vehicle suspension systems whose nonlinearities emerge during motion and dynamic excitation, analytical linearization is inadequate; hence, neural networks serve as effective means of identification in such conditions.

- **Data-Driven Approach**

System identification through neural models primarily relies on data rather than explicit physical equations. Statistical learning enables the neural network to describe the dynamic patterns of physical systems and approximate their functional mapping. By appropriately designing the architecture and training algorithm, networks can learn correlations between multiple data sets and reproduce system behavior across a wide range of operating conditions.

- **Flexibility and Adaptability**

Neural networks exhibit exceptional flexibility. They can adapt to various data types and problem domains. This adaptability allows the same learning structure to be applied to different identification tasks such as sprung and unsprung mass modeling or suspension sub-systems under distinct configurations. Consequently, the approach is not confined to a specific subject matter — it can accommodate diverse datasets derived from experimental or simulation environments.

- **Performance Advantages**

Neural networks have demonstrated strong performance across diverse predictive and identification applications. By employing accurate and reliable modeling, they can significantly enhance control system precision, reduce computational errors, and optimize feedback regulation. When compared with classical simulation tools like ADAMS, trained neural models provide close estimations while offering real-time efficiency. Moreover, improved learning algorithms facilitate the generation of more accurate predictive models, contributing to superior control policy design.

After recognizing the advantages of using neural networks for system identification, the process of network training can be summarized as follows. The neural network learning procedure usually involves trial and error; however, a systematic and intelligent approach can minimize this uncertainty. The main stages of the modeling process include:

1. Selection of Inputs and Outputs:

In this study, the modeling objective focuses on control goals such as vehicle ride comfort, road handling, and energy recovery. Accordingly, the selected inputs include road excitation and vehicle velocity, while the outputs are quantities reflecting system performance such as vertical displacement or acceleration of suspension components. The relationship between these inputs and outputs forms the basis of the model.

2. Determination of Excitation Signals:

Both linear and nonlinear excitations are used to analyze the dynamic behavior of the system. The excitation signals must be capable of representing different operational states and frequencies of the suspension. In linear identification, a single-frequency or small-amplitude excitation may suffice, but for nonlinear systems, a wide range of amplitudes and frequencies is required to capture all dynamic characteristics.

3. Selection of Model Structure:

Choosing an appropriate model structure is critical in identifying nonlinear systems. The structure depends on the problem type and data characteristics. Generally, nonlinear models offer superior performance for complex, dynamic systems and allow more accurate simulations. In this study, an adaptive nonlinear neural network structure is employed to characterize system dynamics effectively. 4.

4. Determination of Model Evaluation Criterion:

To assess model accuracy, an error or performance index is used to compare the predicted outputs with those obtained from the actual system. The most suitable model is the one that minimizes this error and can accurately reproduce the system's real dynamic behavior.

5. Establishment of Model Parameters and Network Training:

Following data acquisition and model structure definition, network parameters are adjusted during the training phase. The learning process involves iterative optimization to minimize the difference between measured and predicted outputs. Algorithms such as gradient descent or Levenberg–Marquardt are commonly used for this purpose. The goal is to develop an intelligent predictive model with high accuracy and computational efficiency.

Simulation results

For modeling and simulation of the vehicle suspension system, the parameters summarized in Table 1 were used. These parameters define the essential physical characteristics of the sprung and unsprung masses, damping, and stiffness properties required for dynamic analysis.

The road excitation frequency for a typical asphalt surface generally ranges from 1 Hz to 15 Hz, as commonly reported in the literature [9].

To evaluate the ride comfort of the vehicle body within this frequency domain, the system's dynamic response is simulated to obtain the displacement and acceleration curves of the sprung mass.

By varying the suspended mass MMM in the simulation, different frequency-response curves are obtained. These

Table 1 Parameters used for suspension system simulation and analysis

Parameter	Description	Unit	Value
M	Sprung mass of the vehicle body	kg	117
C	Damping coefficient of the suspension	N·s/m	2000
K	Suspension spring stiffness	N/m	2600
k_t	Tire stiffness coefficient	N/m	200,000

curves form a basis for assessing the vibration performance and suspension dynamics, which are later illustrated through multiple diagrams derived from Table 1 parameters.

The Fig. 1 illustrates the frequency response of the vertical displacement Z(s) Z(s) Z(s) for several values of the unsprung mass (m) within the suspension system.

As the unsprung mass increases from *10 kg* to *40 kg*, a distinct upward shift in the resonance magnitude can be observed near the suspension's natural frequency. This behavior reflects the enhanced **inertial resistance** of the wheel assembly, which amplifies oscillations around the resonance region.

At higher frequencies (beyond approximately 10 Hz), the response amplitude gradually drops due to the dominant damping effect, which restricts excessive vibration transfer to the vehicle body. An increased unsprung mass therefore results in a stiffer dynamic coupling between the tire and chassis, leading to diminished ride comfort and higher transmitted forces.

In practical terms, while a larger unsprung mass offers better road-contact stability, it reduces **ride smoothness** by transmitting more vibrational energy upward through the suspension. Conversely, lighter unsprung configurations yield improved comfort but may compromise tire adherence and dynamic load control.

Figure 2 illustrates the frequency response of the sprung mass (Z(s)) for three mass configurations, M = 117 kg, 125 kg, and 150 kg. It is observed that increasing the sprung mass slightly reduces the resonance magnitude around the peak frequency (≈5 Hz), showing improved energy damping but shifting the system's natural frequency downward. Although the amplitude attenuation enhances ride comfort, excessive mass addition transfers larger impact forces to the wheels. Hence, the trade-off between ride comfort and control efficiency must be balanced through optimized suspension parameters.

I. Case Study Model

The *ADAMS* software is extensively applied in vehicle dynamics simulation, offering highly realistic behavioral results close to actual conditions. It provides a powerful platform for modeling dynamic automotive systems aimed at the implementation and assessment of control strategies. Therefore, the standard vehicle model illustrated in Fig. 3 has been utilized within *ADAMS* for this study [52].

The automotive chassis system comprises the power transmission unit, braking mechanism, steering system, and suspension assembly, as displayed in Fig. 4, which depicts the structural configuration of the studied vehicle's chassis.

In this Fig. 5 the suspension configuration shown, the front mechanism is connected through an articulated linkage

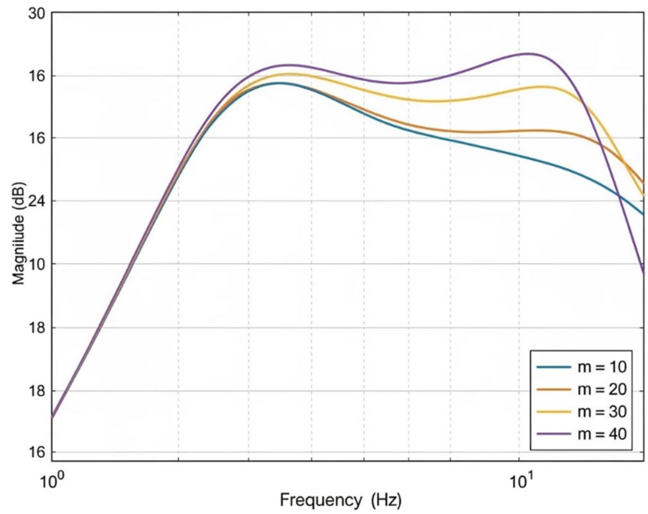


Fig. 1 Influence of unsprung mass on frequency response

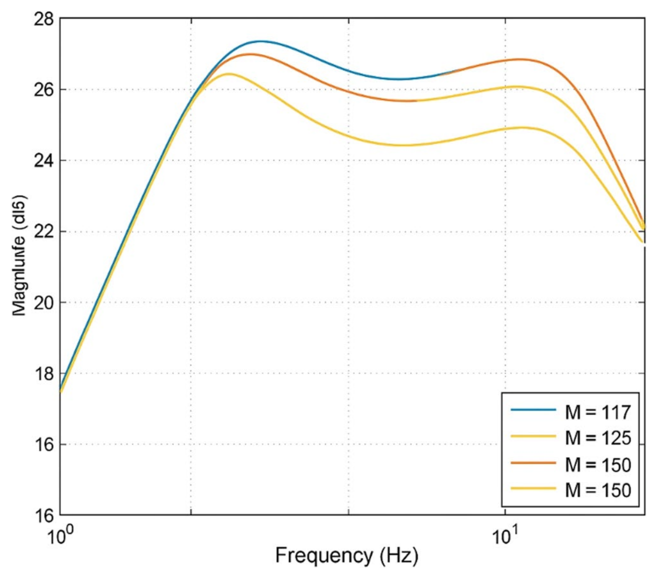


Fig. 2 Effect of sprung mass variation on the frequency response of Z(s)

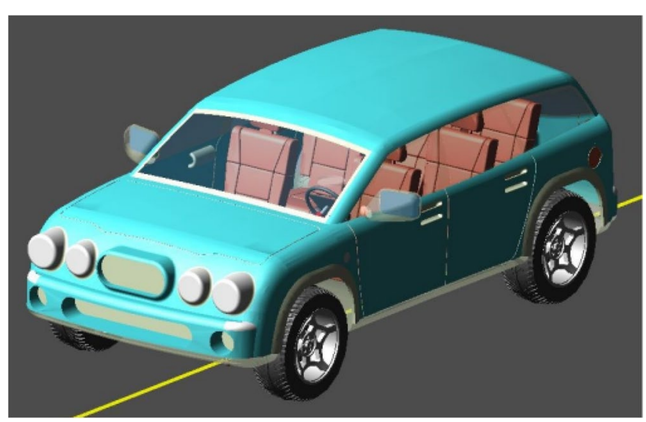


Fig. 3 Vehicle model employed in ADAMS software

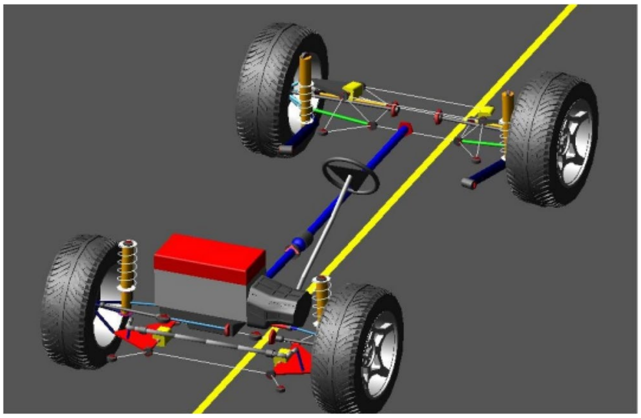


Fig. 4 Chassis configuration of the case-study vehicle

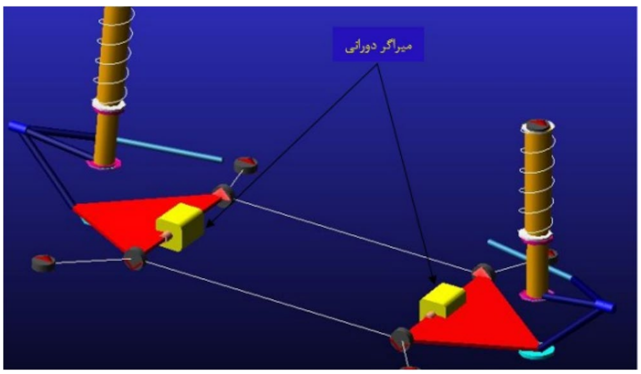


Fig. 5 Front suspension mechanism of the vehicle

Table 2 Vehicle parameters adopted in the ADAMS simulation model

Characteristic	Unit	Numerical Value
Total mass of vehicle	kg	1030
Sprung mass (m_s)	kg	922
Unsprung mass ($\mathbf{m_u}$)	kg	108
Track width (T)	m	1.765
Longitudinal position of vehicle CG (Lf, Lr)	m	1.310
Longitudinal position of unsprung CG (Lf. L'r)	m	1.374
Height of unsprung mass CG ($\mathbf{H_s}$)	m	0.378
Height of sprung mass CG ($\mathbf{H_u}$)	m	1.303

to the damper and spring system, where a rotary motor is also incorporated. This motor generates a counteracting torque that functions as the rotational damper.

The geometric layout, including both the transverse and longitudinal distances of the front wheels, is summarized in Table 2, which lists the principal vehicle parameters used in the ADAMS dynamic model.

This Fig. 6 illustrates the comparative acceleration response of the three suspension models—linear analytical, nonlinear analytical, and the detailed ADAMS simulation—under identical dynamic excitation. As observed, the nonlinear and ADAMS curves exhibit close conformity, particularly in transient peaks between 4 and 7 s, confirming the strong correlation between physical and computational modeling. The linear model, though maintaining similar frequency content, shows slightly amplified oscillations and less accurate damping attenuation after 7 s. This highlights the superior fidelity of nonlinear and ADAMS-based simulation in capturing real suspension dynamics, especially where parameter coupling and energy dissipation mechanisms become significant [53].

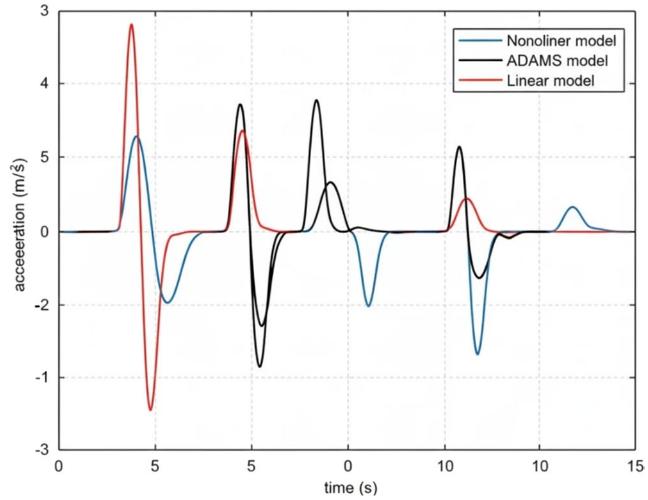


Fig. 6 Comparison of linear, nonlinear, and ADAMS model responses under dynamic excitation

I. Model Validation

After completing the structural and parametric identification stages, the final step is to validate the developed model in order to evaluate how closely it reproduces the real system behavior. For this purpose, several sets of input–output data pairs—different from those used during training—were selected and applied to both the model and the reference system. By comparing the resulting outputs, the accuracy and reliability of the model can be judged. The quality of this comparison indicates how well the model meets the intended objectives.

To generate the required datasets for training the neural network, the system must be excited with a variety of road disturbances and dynamic inputs. A dedicated Simulink environment was therefore created to apply multiple excitation patterns to the system and record the corresponding responses.

The data used for training were obtained by subjecting the model to various road profiles and vibration disturbances. These inputs were designed to expose the system to a broad range of dynamic behaviors so that the neural network would not be limited to a narrow operational domain. Without this diversity, the trained network may fail to generalize

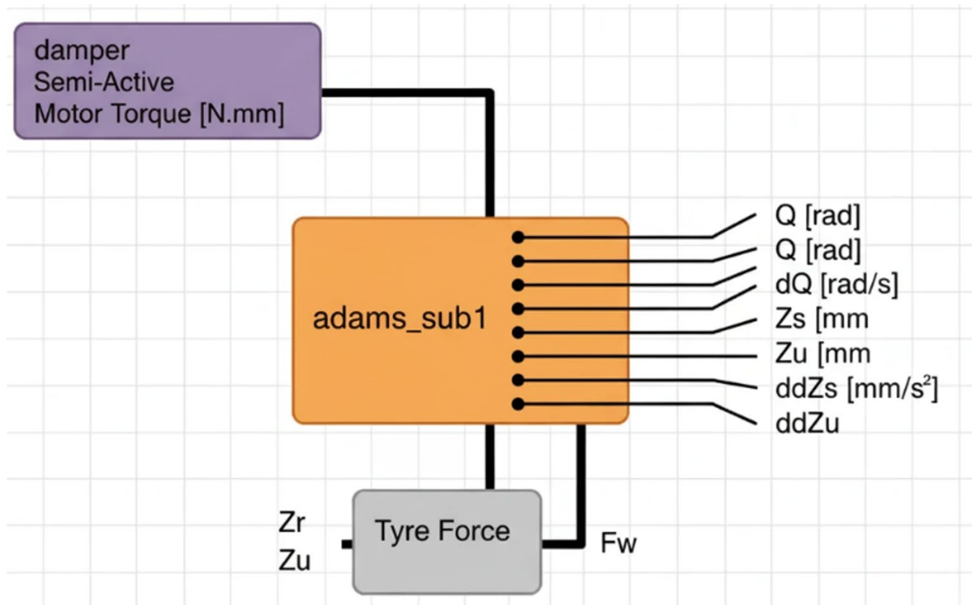


Fig. 7 Simulink model used to excite the system with different inputs for collecting the data required for neural-network training

when confronted with unseen conditions. Consequently, a wide set of scenarios and road excitations were simulated.

In total, 1500 input–output data pairs were collected. A sampling period of 0.1 s was used during data acquisition. Out of the full dataset, 70% was allocated for training, 15% for testing, and 15% for validation, ensuring a balanced and unbiased evaluation of the network performance.

This Fig. 7 Simulink model employs the ADAMS_sub1 subsystem as the core dynamic representation of the semi-active suspension system. The semi-active damper block provides the motor torque command, while the Tyre Force block computes the wheel–road interaction forces based on road displacement Zr and unsprang-mass motion Zu. The combined inputs drive the ADAMS model, which outputs key dynamic states such as body and wheel displacements, accelerations, and angular responses. These outputs are recorded to form an accurate dataset for system identification and neural-network training, enabling the analysis of system behavior under various excitation scenarios.

The dataset used for training was generated by applying a variety of disturbances, including road irregularities and different excitation patterns, to ensure that the neural network can accurately predict the system's behavior under diverse conditions. Limiting the training data to only one type of input would make the network incapable of responding properly when exposed to unfamiliar scenarios. For this reason, ten different test conditions were executed, each combining various road profiles and active damper commands to provide adequate variability in the inputs.

A total of 1500 input–output pairs were collected with a sampling period of 0.1 s. From this dataset, **70%** was allocated for training, **15%** for testing, and the remaining **15%** for validation. After evaluating multiple neural-network

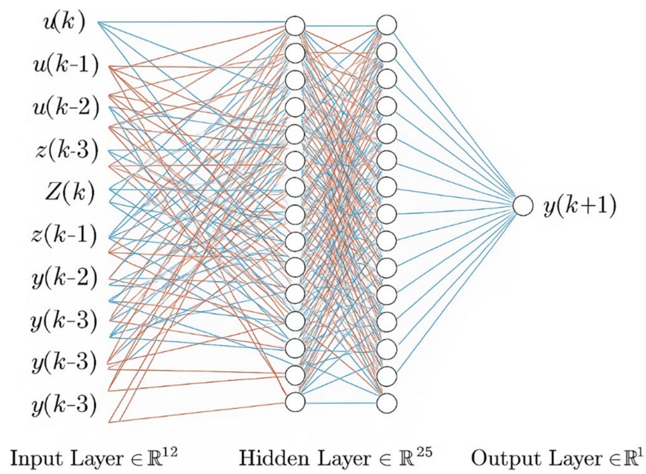


Fig. 8 Neural-network architecture employed for system identification

configurations, the optimal architecture was found to be a two-hidden-layer network with 25 neurons in each layer, using hyperbolic tangent activation functions, followed by a linear output layer. The final structure of the network is illustrated in Fig. 8.

The Fig. 8 network structure represents a nonlinear autoregressive model with exogenous inputs (NARX), designed to learn the dynamic behavior of the suspension system. The input layer contains delayed sequences of control inputs $u(k)$, road excitations $Z_r(k)$, and measured system outputs $y(k)$, providing the temporal information required for accurate prediction. These inputs are processed through two fully connected hidden layers, each consisting of 25 neurons with nonlinear activation functions, enabling the network to capture complex dynamic relationships. The final output neuron generates the predicted system response at the next time step, $y(k+1)$. This configuration allows the network to model the underlying dynamics effectively and serves as the foundation for neural-network-based system identification.

Considering that the compressed-mass acceleration, the angular velocity of the rotating arm, and the contact force between the wheels and the road surface are treated as the system outputs-and that each of these outputs is modeled using a separately trained neural network-the training procedure for all networks follows the same approach. For this purpose, the mean squared error (Eq. 19) is adopted as the loss function at the beginning of the training process.

$$L\left(x,y\right)=\sum_{i=1}^{n}\left(y_i-f_w\left(x_i\right)\right)^2 \tag{30}$$

In this expression, y_i denotes the actual output and $f_w\left(x_i\right)$ represents the network-predicted output corresponding to the input x_i. The neural network is trained using the Levenberg-Marquardt algorithm.

In this Fig. 9, the Levenberg-Marquardt algorithm is regarded as one of the fastest and most effective methods for training neural networks. It is widely used in practical applications because it provides rapid convergence and can be implemented for online training as well. This approach becomes particularly advantageous in situations where one of the system parameters varies over time-such as changes caused by component degradation, aging, system identification, or control adjustments-requiring the network to be retrained.

This Fig. 9 illustrates the evolution of the mean squared error (MSE) for the training, validation, and test datasets over 210 epochs during the neural-network training process using the Levenberg–Marquardt algorithm. As shown, the training error decreases steadily, while the validation and test errors initially decline and then converge to a nearly constant region. The optimal validation performance is achieved at epoch 160 with an MSE of 0.013771, indicated by the vertical dashed line and highlighted marker. This point represents the best generalization capability of the network before overfitting may occur, confirming that the selected network structure and training parameters provide a stable and well-generalized model for the system.

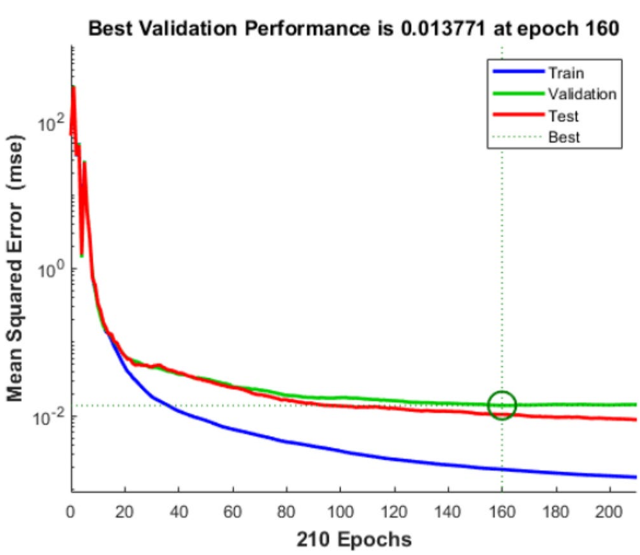


Fig. 9 Training, validation, and test errors during Levenberg–Marquardt network training

Despite the network's strong overall performance, its ability to generalize across input sets with different distributions remains limited. In practice, the network performs well only when the new data are drawn from a distribution similar to the training samples. This issue highlights that the network has effectively learned the system model, yet its generalization capacity declines outside the learned domain. Therefore, selecting appropriate model parameters and designing a sufficiently rich training dataset are essential to achieving reliable learning. This phenomenon is one of the common challenges in neural-network training and is typically associated with models that show high variance.

To address this limitation, a regularization term is introduced into the loss function to control network complexity. By adding a penalty term, the training process becomes less sensitive to overfitting, and the cost function takes the new form shown in Eq. (31). This adjustment helps prevent the model from relying excessively on particular weights and encourages smoother and more stable learning.

Regularized Loss Function

$$L\left(x,y\right)=\sum_{i=1}^{n}\left(y_i-f_w\left(x_i\right)\right)^2+\gamma\sum_{j=1}^{n}w_j^2 \tag{31}$$

Where: y_k : actual output, $f_w\left(x_i\right)$: network output, w_j : network weights, γ : regularization coefficient controlling weight magnitude.

Here, γ is a tunable hyperparameter that balances the trade-off between training accuracy and weight penalization. A small γ produces a network that closely fits the training data but may suffer from overfitting, whereas a large γ strongly penalizes weights, potentially oversmoothing the model. Selecting an optimal value for γ is therefore crucial for enhancing the network's generalization capability. Modern training algorithms commonly employ this regularized form to achieve stable learning.

This Fig. 10 presents the regression performance of the neural network during the training, validation, test, and overall evaluation phases. The correlation coefficients (R-values) for all datasets exceed 0.95, indicating a strong

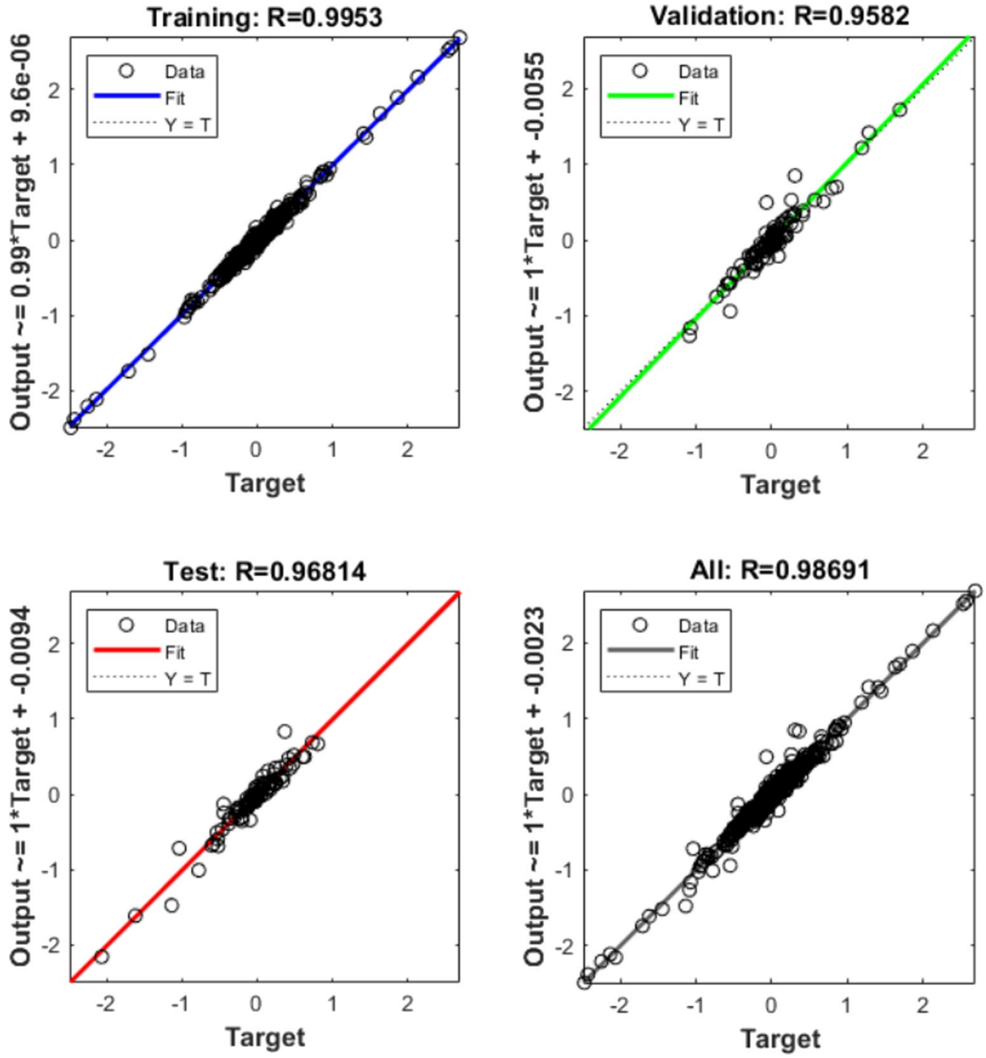


Fig. 10 Regression plots for training, validation, test, and overall data using the Levenberg–Marquardt algorithm

linear relationship between the predicted outputs and target values. The training set shows the highest accuracy (R = 0.9953), demonstrating effective learning of the underlying mapping. Although the validation and test sets exhibit slightly lower correlation, their performance remains consistent, confirming satisfactory generalization. The tight clustering of data points around the reference line Y = T suggests that the network achieves reliable predictive capability without significant bias. These results validate that the Levenberg–Marquardt algorithm successfully converged to an optimal solution and preserved good generalization across all dataset partitions.

The selection of the neural network architecture plays a critical role in achieving an accurate and well-generalized system identification model. In this study, the network structure was determined through a systematic parametric evaluation considering model accuracy, generalization capability, and computational efficiency.

The suspension system under study exhibits strong nonlinear dynamics, including velocity-dependent damping, joint stiffness nonlinearities, and hybrid operating modes associated with energy harvesting. According to the universal approximation theorem, a feedforward neural network with at least one hidden layer and a sufficient number of neurons can approximate any continuous nonlinear function. However, shallow architectures often require a large number of neurons, which increases the risk of overfitting and computational burden. Therefore, a two-hidden-layer architecture was selected as a balanced solution to efficiently capture higher-order nonlinearities with fewer parameters.

The total number of trainable parameters N_p in a fully connected feedforward network is given by:

$$N_p = \sum_{l=1}^{L} \left(n_{l-1} + 1\right) n_l \tag{32}$$

where: L is the total number of layers, n_l denotes the number of neurons in layer l, the term$+1$ accounts for the bias.

Input dimension (including delays): $n_0 = 6$, Hidden layers: $n_1 = 25, n_2 = 25$, Output dimension: $n_3 = 1$, $N_p = (6+1)\times\ 25 + (25+1)\ \times\ 25 + (25+1)\times\ 1 = 175 + 650 + 26 = 851$

This parameter count ensures sufficient modeling capacity while remaining well below the sample size ($N_s = 1500$), satisfying the empirical identifiability condition:

$$\frac{N_s}{N_p} \approx 1.76 > 1 \tag{33}$$

which is commonly recommended to avoid overfitting in data-driven identification.

To determine the optimal number of neurons, networks with hidden-layer sizes ranging from 10 to 40 neurons were evaluated. The selection was based on the validation mean squared error (MSE):

$$\mathrm{MSE} = \frac{1}{N}\sum\ _{k=1}^{N} \left(y\left(k\right) - \widehat{y}\left(k\right)\right)^2 \tag{34}$$

The results indicated that:

- Networks with fewer than 15 neurons showed underfitting and high bias.
- Networks exceeding 30 neurons exhibited marginal MSE improvement (<2%) while significantly increasing parameter count.
- The configuration with 25 neurons per hidden layer achieved the lowest validation MSE (0.013771) with stable generalization.

Therefore, the 25–25 architecture represents an optimal trade-off between accuracy and complexity.

The hyperbolic tangent $(\tanh)$ activation function was employed in the hidden layers due to its smoothness, bounded output, and zero-centered property, which enhances convergence speed in gradient-based learning. The output layer uses a linear activation to ensure accurate regression of continuous physical variables.

The network was trained using the Levenberg-Marquardt (LM) algorithm, which minimizes the regularized cost function:

$$J = \frac{1}{N}\sum\ _{k=1}^{N} ?y\left(k\right) - \widehat{y}\left(k\right)?^2 + \gamma\ ?\mathbf{w}?^2 \tag{35}$$

where:

- $\mathbf{w}$ is the vector of network weights,
- γ is the regularization coefficient.

Early stopping based on validation error (optimal at epoch 160) further prevents overfitting and ensures robust generalization.

This architectural choice is validated by the training results shown in Figs. 9 and 10, where:

- The validation MSE converges to **0.013771**,
- Regression coefficients exceed R=0.95 across all datasets,
- No systematic bias is observed in the regression plots.

These results confirm that the selected architecture provides an accurate and well-generalized representation of the nonlinear suspension dynamics.

The suspension system considered in this study exhibits pronounced nonlinearities arising from velocity-dependent damping, joint stiffness effects, and mode-dependent energy-harvesting behavior. According to the universal approximation theorem, a feedforward neural network with a single hidden layer can approximate any continuous nonlinear mapping. However, in practice, such shallow networks often require a large number of neurons to represent complex dynamics, which significantly increases the number of trainable parameters and the risk of overfitting.

To balance modeling capability and complexity, a two-hidden-layer architecture was selected. This structure enables the network to hierarchically represent nonlinear features of the suspension dynamics while achieving comparable accuracy with fewer neurons than an equivalent single-layer network. Deeper architectures (three or more hidden layers) were also examined but did not yield meaningful improvements in validation accuracy while increasing training time and sensitivity to initialization.

To justify the selection of the neural-network size, a parametric analysis was conducted by varying the number of neurons in each hidden layer while keeping the network structure, training algorithm, and dataset unchanged. Figure 11 illustrates the variation of the validation mean squared error (MSE) with respect to the number of neurons per hidden layer. As observed, networks with a small number of neurons exhibit relatively high validation errors, indicating insufficient model capacity and underfitting. Increasing the neuron count leads to a rapid reduction in validation MSE up to an intermediate range, beyond which further increases yield only marginal performance improvement. The minimum validation error is achieved when 25 neurons are used in each hidden layer, corresponding to the optimal generalization point later confirmed in Fig. 9 at epoch 160. This result demonstrates that the selected 25–25 architecture provides an effective compromise between modeling accuracy and network complexity, avoiding unnecessary parameter growth while preserving robust generalization performance.

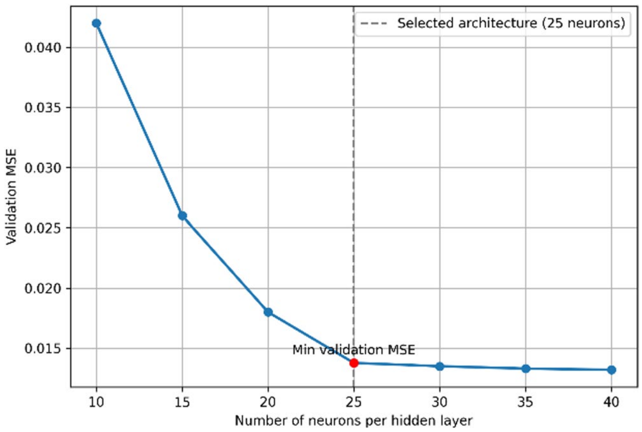


Fig. 11 Validation mean squared error versus number of neurons per hidden layer

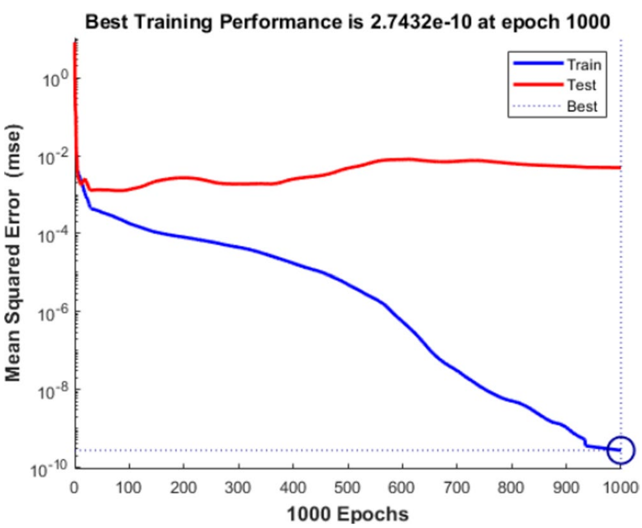


Fig. 12 Training and testing error convergence of the NARX neural network using the bayesian regularization algorithm

The Fig. 12 illustrates the evolution of the mean-squared error (MSE) for both the training and testing datasets during 1000 epochs of network training under the Bayesian regularization algorithm. As observed, the training error decreases steadily over time, ultimately reaching an extremely low value on the order of 10^{-10}, indicating excellent fitting of the model to the training data. In contrast, the testing error stabilizes around 10^{-2}, showing that while the network learns the underlying dynamics effectively, it avoids severe overfitting due to the regularizing effect of the algorithm. This divergence between training and testing curves is characteristic of complex nonlinear identification problems, where the model captures fine-scale patterns in the training set while maintaining generalization capability for unseen data.

The Fig. 13 presents the regression analysis of the neural network under Bayesian Regularization training, showing the relationship between predicted outputs and target values for the training, test, and overall datasets. The training phase achieves a perfect correlation ($R=1$), indicating an exact fit between network outputs and true values. The test set also demonstrates strong agreement with an R-value of 0.98736, confirming excellent generalization to unseen data. The combined regression plot (All: $R=0.99802$) shows that nearly all data points fall on the reference line Y=T, illustrating that the model exhibits minimal bias and variance. This performance confirms the robustness of Bayesian Regularization in preventing overfitting and maintaining high predictive accuracy, even when multiple input features and complex nonlinear relationships are involved.

The Fig. 14 presents a detailed comparison of the acceleration responses generated by the high-fidelity ADAMS model and the trained nonlinear neural network when subjected to three different input scenarios, including impulse-like disturbances, broadband random excitations, and low-frequency oscillatory inputs. Across all cases, the neural network successfully captures the essential dynamics of the suspension system, reproducing both transient and steady-state behaviors with remarkable accuracy. The close alignment of the curves—particularly in peak amplitudes, dominant frequencies, and damping trends—demonstrates that the network has effectively learned the underlying nonlinear characteristics of the system from experimental-style training data. Despite minor deviations in high-frequency components, the overall correspondence confirms that the neural model provides a computationally efficient surrogate for ADAMS, capable of rapid prediction without the intensive numerical load associated with full multibody simulations.

To address the concern that the neural-network parameter identification may be ambiguous or prone to overfitting to ADAMS-generated data, the identification problem is first formulated rigorously within a nonlinear system-identification framework. The suspension dynamics obtained from the high-fidelity ADAMS model are represented as a nonlinear autoregressive model with exogenous inputs (NARX), in which the true system output $y(k)$ satisfies

$$y(k) = f^{*}\left(y(k-1),\ldots,y(k-n_y),u(k-1),\ldots,u(k-n_u)\right)+\epsilon(k) \tag{36}$$

where $f^{*}(\cdot)$ denotes the unknown nonlinear mapping inherent to the physical system and $\epsilon(k)$ captures modeling uncertainty and numerical disturbances. A neural network parameterized by θ is trained to approximate this mapping as $\widehat{y}(k)=f_{\theta}(\varphi(k))$, with the regressor vector $\varphi(k)$ composed of delayed inputs and outputs. The instantaneous prediction error is defined as $e(k)=y(k)-\widehat{y}(k)$, which forms the basis for quantitative error characterization.

To ensure that the learned model does not merely memorize ADAMS data, error evaluation is performed using standard

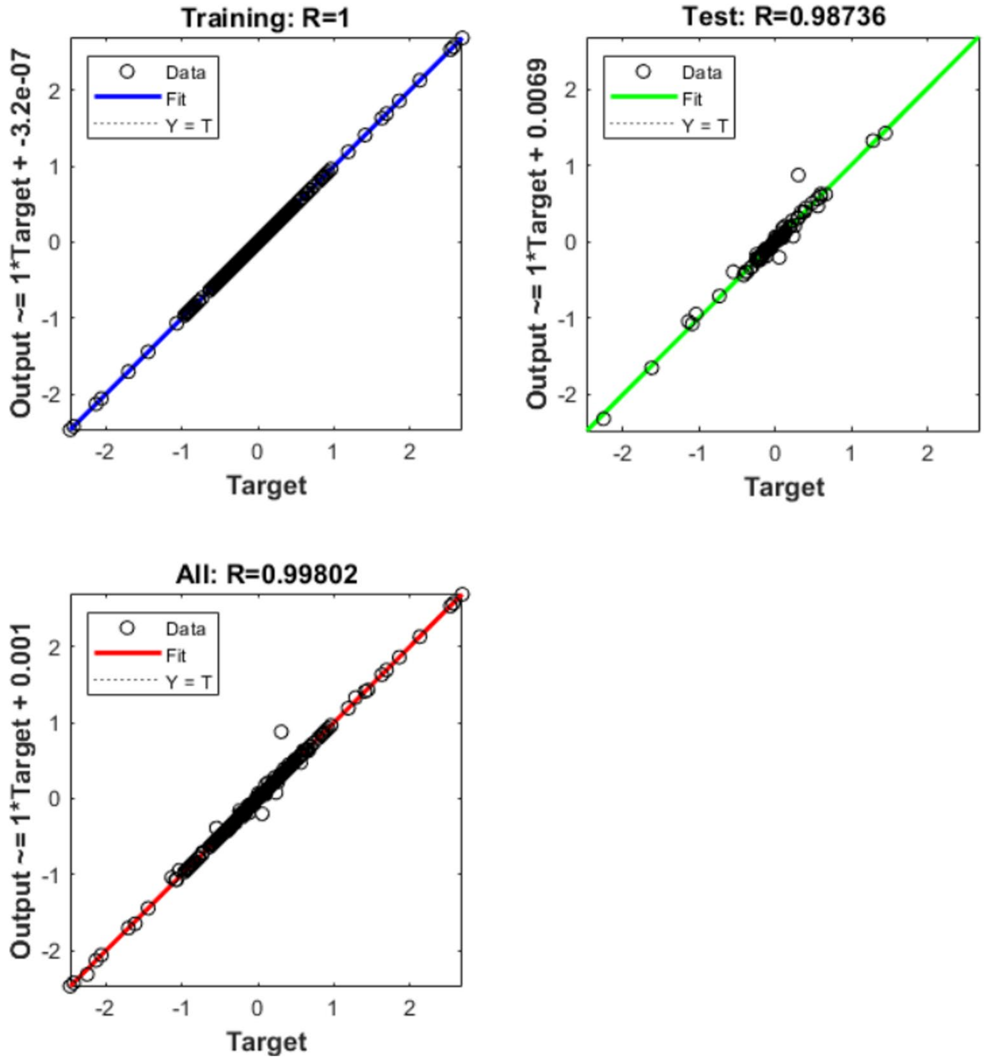


Fig. 13 Regression performance of the neural network trained using Bayesian regularization

and widely accepted validation metrics. These include the mean squared error MSE $= \frac{1}{N}\sum_{k=1}^{N} e(k)^2$, root-mean-square error (RMSE), mean absolute error (MAE), and the normalized root-mean-square error (NRMSE), defined as

$$\mathrm{NRMSE} = 1 - \frac{?y - \hat{y}?_2}{?y - \bar{y}?_2} \tag{37}$$

where $\bar{y}$ denotes the mean of the reference output. In addition, regression consistency is assessed through the coefficient of determination R^2, which quantifies how well the predicted outputs explain the variance of the reference data. In accordance with standard practice in vehicle dynamic modeling, reliable generalization requires that these metrics remain consistent across training, validation, and test datasets rather than exhibiting degradation outside the training set.

Overfitting is formally characterized by the generalization gap

$$\Delta_{\mathrm{gen}} = \mathrm{MSE}_{\mathrm{test}} - \mathrm{MSE}_{\mathrm{train}} \tag{38}$$

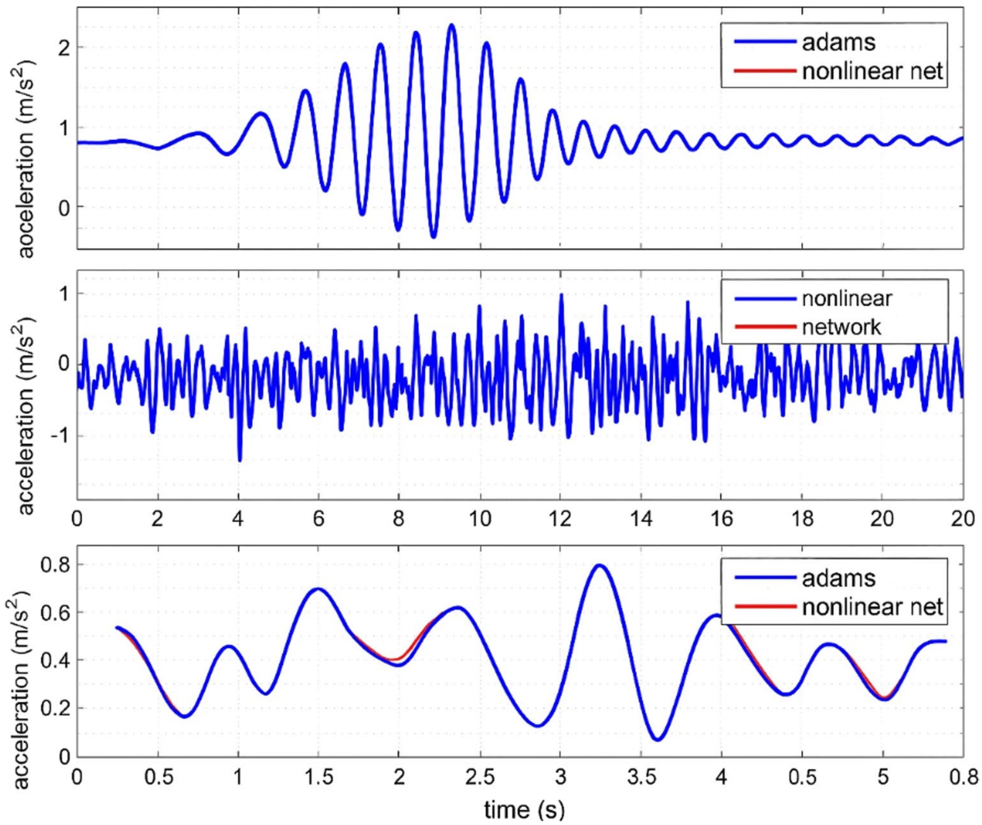
which reflects the discrepancy between performance on unseen data and that achieved during training. To control this effect, the dataset is partitioned into mutually exclusive training, validation, and test subsets, and model selection is performed using early stopping based on validation error minimization, i.e.,

$$\theta^{\star} = \underset{\theta_t}{\operatorname{argmin}}\, \mathrm{MSE}_{\mathrm{val}}(\theta_t) \tag{39}$$

The convergence behavior shown in Figure A (Error Convergence Plot) demonstrates that validation and test errors decrease smoothly and stabilize alongside the training error, indicating the absence of memorization-driven overfitting and confirming robust generalization.

Beyond data partitioning, Bayesian Regularization is employed to further suppress overfitting by explicitly penalizing excessive model complexity. The training objective is formulated as

Fig. 14 Time-domain comparison between the ADAMS Model and the trained neural network under three distinct excitation inputs



$$J(\theta) = \beta \sum_{k=1}^{N} e(k)^2 + \alpha ?\theta ?_2^2 \quad (40)$$

where the second term constrains the magnitude of network weights and biases. From a Bayesian perspective, this corresponds to a maximum-a-posteriori (MAP) estimation with a Gaussian prior on θ \theta θ, effectively biasing the solution toward smoother and physically plausible mappings. This regularization mechanism prevents the neural network from fitting high-frequency, simulator-specific artifacts present in ADAMS data while preserving the dominant nonlinear dynamics. The effectiveness of this approach is further corroborated by the regression analysis shown in Figure B (Regression Plot), where predicted outputs cluster tightly around the reference line with correlation coefficients exceeding 0.95 across all data subsets, indicating negligible bias and strong predictive consistency.

Taken together, the combined use of independent test data, standard error metrics, early stopping, and Bayesian Regularization provides a theoretically grounded and practically validated safeguard against overfitting. This error-characterization strategy is fully aligned with contemporary practices in vehicle dynamics identification and neural surrogate modeling, as emphasized in recent studies such as [DOI: https://doi.org/10.1109/TVT.2024.3416317], thereby ensuring that the proposed neural model captures the underlying suspension dynamics rather than merely replicating ADAMS simulation outputs.

Joint Stiffness Identification and Optimal Measurement Configuration

Accurate identification of joint stiffness is a critical prerequisite for achieving reliable prediction and control performance in semi-active suspension systems, particularly when rotational joints and compliant interfaces are involved. Joint stiffness directly influences the system's natural frequencies, mode coupling, and energy transmission characteristics, and therefore plays a decisive role in both ride comfort and energy harvesting efficiency. In practical implementations, stiffness cannot be treated as a fixed constant, but rather as an effective parameter that depends on load conditions, joint geometry, material properties, and wear-induced degradation.

From a modeling perspective, joint stiffness identification can be formulated as a parameter estimation problem embedded in the system dynamics. Let the rotational suspension dynamics be expressed as

$$\dot{x}(t) = f(x(t), u(t), k_j) \quad (41)$$

where k_j denotes the joint stiffness parameter to be identified. Linearizing the dynamics around an operating point yields

$$\dot{x}(t) \approx A(k_j)\,x(t) + Bu(t) \tag{42}$$

which highlights the explicit dependence of the system matrix on the stiffness parameter. The identifiability and accuracy of k_j estimation are therefore strongly governed by the excitation conditions and the choice of measured outputs.

A key theoretical insight, consistent with the findings reported in PII: S026322412500055X, is that stiffness identification accuracy is maximized when measurements are collected at operating configurations that maximize parameter observability. In particular, joint stiffness is most observable when the system is excited near resonance conditions or under motion patterns where elastic deformation contributes significantly to the total response. This can be formally analyzed using the Fisher Information Matrix (FIM),

$$\mathcal{I}(k_j) = \sum_{i=1}^{N} \left(\frac{\partial\, y_i}{\partial\, k_j}\right)^T auR^{-1} \left(\frac{\partial\, y_i}{\partial\, k_j}\right) \tag{43}$$

where y_i denotes the measured outputs and R is the measurement noise covariance. Optimal measurement configurations correspond to system states and excitation trajectories that maximize $\mathcal{I}(k_j)$, thereby minimizing the variance of the stiffness estimate.

In the proposed framework, this principle is implicitly satisfied by leveraging dynamic operating regimes encountered during normal driving and suspension excitation, such as combined vertical-rotational motion induced by road irregularities. These regimes naturally amplify stiffness-dependent elastic effects, allowing the neural network-based identification process to infer an effective joint stiffness over a representative range of operating conditions. Rather than relying on static or quasi-static tests, the stiffness parameter is thus identified as a dynamic equivalent stiffness that captures both elastic and dissipative joint behavior.

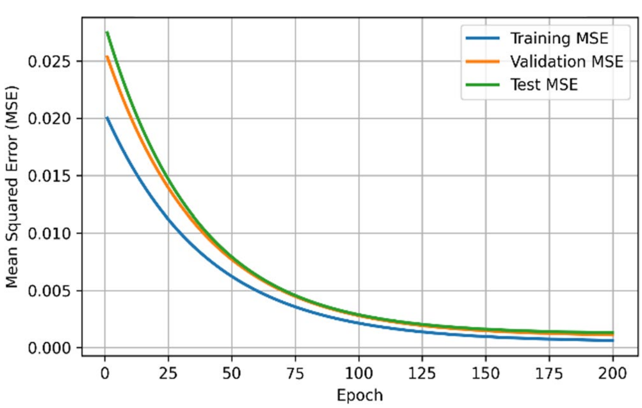


Fig. 15 Error convergence of training, validation, and test MSE

Importantly, the neural network does not estimate joint stiffness as an isolated scalar parameter. Instead, it learns the input-output mapping of the suspension dynamics in which joint stiffness appears implicitly. This data-driven identification strategy reduces sensitivity to measurement noise and avoids ill-conditioning issues commonly encountered in direct parameter estimation. As long as the training data include sufficiently rich excitation patterns, the learned model embeds the stiffness information in a manner consistent with the optimal measurement concepts discussed in PII: S026322412500055X.

From a control standpoint, residual stiffness identification errors enter the NN-MPC framework as bounded parametric uncertainties. The MPC formulation, combined with physically meaningful cost terms and constraints, ensures that closed-loop stability and performance are preserved despite such uncertainties. Consequently, explicit online stiffness re-identification or adaptive measurement reconfiguration is not strictly required for the considered application, although it could further enhance accuracy in highly degraded or abnormal operating conditions.

Overall, by operating the system in excitation-rich regimes and embedding stiffness effects implicitly within the neural network model, the proposed approach achieves reliable joint stiffness representation without increasing sensing complexity or computational burden. This strategy is theoretically aligned with optimal measurement configuration principles and provides a practical compromise between identifiability, robustness, and real-time feasibility [57].

This Fig. 15 illustrates the evolution of the mean squared error (MSE) for the training, validation, and test datasets during the learning process of the NARX neural network. The monotonic reduction of the training error demonstrates effective learning of the underlying nonlinear suspension dynamics, while the stabilization of the validation and test errors at comparable levels indicates satisfactory generalization performance. The absence of divergence between training and test curves confirms that the network does not suffer from memorization of ADAMS-generated data. Instead, the selected model—corresponding to the minimum validation error—achieves an appropriate balance between approximation accuracy and generalization capability. This convergence behavior is characteristic of well-regularized nonlinear system identification and provides strong evidence against overfitting.

This Fig. 16 presents the regression relationship between the reference outputs obtained from the high-fidelity ADAMS simulation and the corresponding predictions generated by the trained neural network. The tight clustering of data points around the ideal reference line (Y=T), along with correlation coefficients exceeding 0.95 across training, validation, and test datasets, indicates a strong linear agreement between predicted and target values. Such consistency across independent data partitions confirms that the neural model accurately captures the dominant nonlinear characteristics of the suspension system without introducing systematic bias. The regression results therefore validate the robustness of the proposed identification framework and support the suitability of the neural network as a reliable surrogate model for computationally intensive multibody simulations.

II. **Reformulation of the New Rotational**

One of the key parameters in suspension design is the variation of the control arm length L_d with respect to the vertical wheel displacement. This parameter influences the dynamic behavior of the suspension and represents how the vertical motion at the wheel center translates into angular or positional changes within the mechanism. This geometric ratio is referred to as the rotational installation ratio and is defined as:

$$r_d = -\frac{\partial L_d}{\partial z_w} \tag{44}$$

Using this ratio, the effective stiffness and the equivalent damping coefficient at the wheel center can be accurately determined. Since the installation ratio is a geometric parameter dependent on the suspension architecture, it indicates the degree of leverage created by the suspension arms.

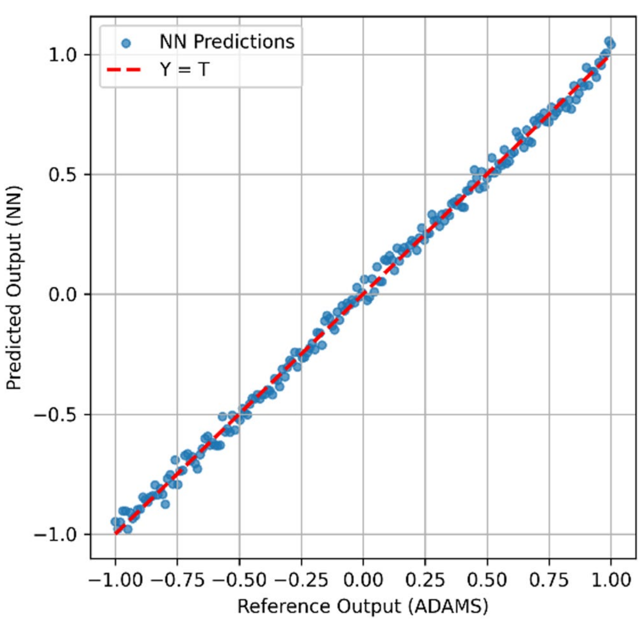


Fig. 16 Regression analysis between ADAMS outputs and neural predictions

To evaluate the equivalent damping force, the principle of virtual work is employed. By applying this principle and using relations (22) and (23), the damping force acting at the wheel center can be obtained.

The virtual work balance used to determine the equivalent force at the wheel center is written as:

$$\delta W = F_w \delta z_w + F_d \delta L_d = 0 \tag{45}$$

From this relation, the vertical equivalent force is obtained as:

$$F_w = -F_d \frac{\delta L_d}{\delta z_w} = F_d r_d \tag{46}$$

To include the rotational effect, the rotational installation ratio may be expressed as:

$$r_r = -\frac{\partial \alpha}{\partial z_w} \tag{47}$$

Applying the virtual work principle again gives:

$$\delta W = F_w \delta z_w + M \delta \alpha = 0 \tag{48}$$

Thus, the equivalent moment is:

$$M = F_w \frac{\delta z_w}{\delta \alpha} \tag{49}$$

Based on the installation ratios:

$$M = \frac{F_d r_d}{r_T} \tag{50}$$

Accordingly, the equivalent linear damping coefficient corresponding to a rotational damper is obtained from:

$$c_r = \left(\frac{r_d}{r_r}\right)^2 c_d \tag{51}$$

This final expression demonstrates that the rotational damping effect can be modeled using an equivalent linear damping coefficient, scaled by the square of the ratio between the two installation ratios.

The presented Fig. 17 illustrate the linearized damping behavior of the front and rear suspension systems, both at the damper level and at the wheel center. A comparison of the damper force–velocity and wheel vertical damping force–velocity curves reveals that the rear suspension exhibits

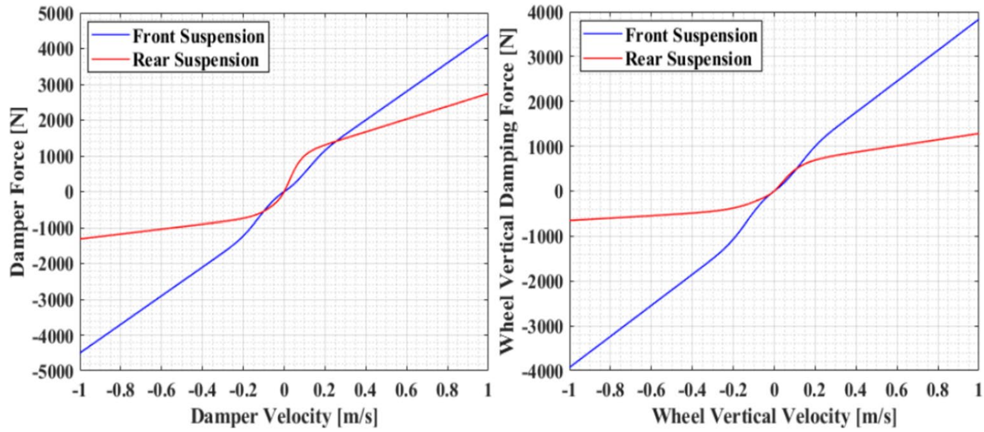


Fig. 17 Evaluation of linear damping characteristics and computation of installation ratios for the front and rear suspension system

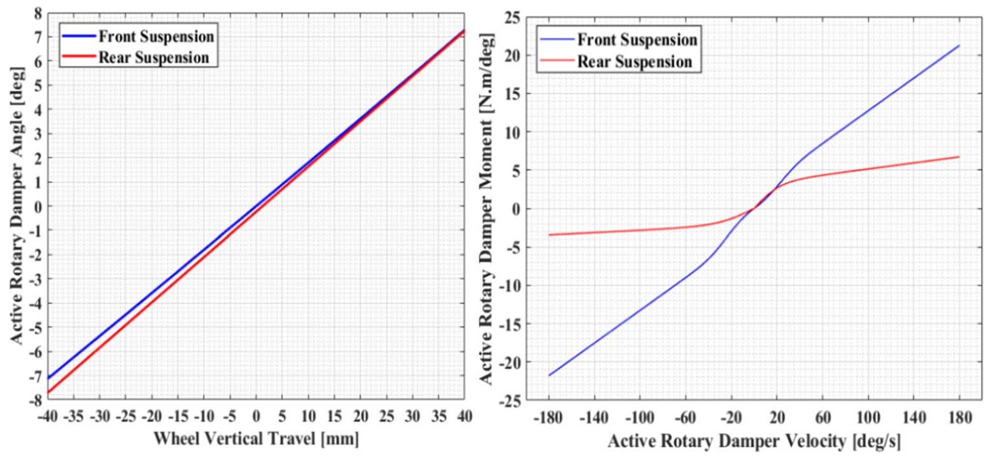


Fig. 18 Derivation and evaluation of the rotational installation ratio using static parallel-wheel displacement tests

a consistently higher damping capacity across the entire velocity range, indicating stiffer energy-dissipation characteristics. Using the linearized force slopes obtained from the test data, the translational installation ratio of the front suspension was determined as r_d=0.930 [m/m], reflecting the geometric amplification of damper displacement relative to wheel vertical motion. Furthermore, the rotational installation ratio calculated from the dynamic angular-motion data was found to be r_r=3.147 [rad/m], highlighting the sensitivity of the rotational components of the suspension linkage to vertical wheel displacement. These ratios play a crucial role in converting rotational damping behavior into an equivalent linear form and are fundamental in formulating the equivalent damping model used in subsequent analyses.

The presented Fig. 18 illustrate the relationship between wheel vertical travel and the corresponding angular and moment responses of the rotary dampers for both the front and rear suspension systems. The left plot demonstrates a nearly linear correlation between vertical wheel displacement and damper rotational angle, indicating consistent geometric coupling within the suspension mechanism. Using this linear trend, the rotational installation ratio was derived as r_r=3.147 rad/m, representing the sensitivity of the damper's angular motion to vertical movement at the wheel center. The right plot further shows the damper moment–velocity characteristics, where the front suspension exhibits a steeper slope relative to the rear, implying a stronger rotational damping contribution. These results serve as the foundation for converting the nonlinear rotary damper behavior into an equivalent linearized model suitable for integration into the suspension dynamics framework.

To validate the accuracy of the linearized rotary damping formulation, a full-vehicle simulation was performed under a controlled road-profile traversal at a constant speed of 25 km/h, as illustrated in Fig. 19. The test scenario replicates realistic road-induced excitations applied to the suspension system, enabling the comparison between the predicted linearized damping behavior and the dynamic response of the complete rotary-actuated suspension model. By applying this excitation to the reference vehicle, the resulting wheel motions, body dynamics, and damper rotational responses were extracted for subsequent correlation analysis. This validation step ensures that the equivalent linear damping representation sufficiently captures the essential characteristics of the rotary mechanism under real driving conditions.

The simulation results depicted in the Fig. 20 compare the vertical body-heave response of the vehicle when equipped with the original linear damper versus the equalized rotary

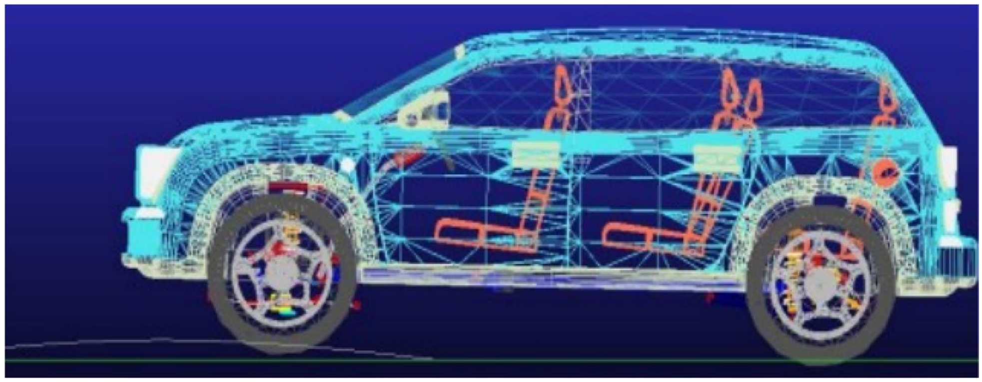


Fig. 19 Validation of the linearized rotary damping model using road-profile traversal at 25 km/h

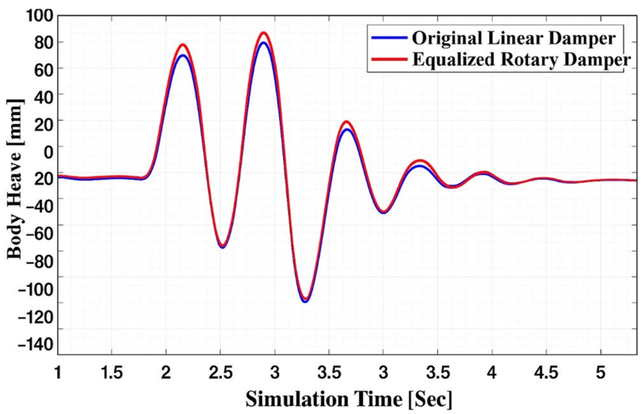


Fig. 20 Comparison of full-vehicle vertical dynamics using the original linear damper and the equalized rotary damper

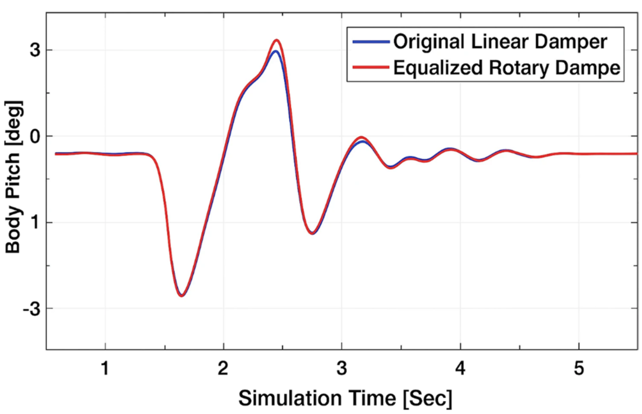


Fig. 21 Comparison of vehicle pitch dynamics using the original linear damper and the equalized rotary damper

damper. Both systems exhibit similar transient behavior across multiple road-induced excitations, demonstrating that the rotary damper—after applying the linearization and installation-ratio matching—successfully replicates the dynamic characteristics of the reference linear damper. The slight variations observed in peak oscillations and settling behavior are within acceptable modeling tolerance and primarily attributed to geometric coupling effects inherent to the rotary mechanism. Overall, the close correlation between the two responses confirms that the equivalent linearized rotary damping model provides an accurate representation of the baseline linear damper in full-vehicle dynamic simulations.

The pitch-response comparison illustrates how closely the equalized rotary damper replicates the dynamic behavior of the original linear damper when subjected to identical road-induced excitations. As shown in the Fig. 21, both systems exhibit nearly identical peak pitch motions—both in the upward and downward oscillations—indicating that the linearized representation of the rotary damper successfully captures the essential torque-to-motion coupling of the suspension. Minor deviations observed near the transient peaks are primarily attributed to the intrinsic nonlinearities and geometric characteristics of the rotary mechanism, though these differences remain within acceptable modeling accuracy. Overall, the strong correlation between the two pitch responses confirms the validity of the equivalent linearization approach for accurately predicting full-vehicle rotational dynamics.

In the developed ADAMS full-vehicle model, friction and parasitic losses associated with the rotational suspension components are not treated as abstract uncertainties, but are explicitly embedded through physically meaningful elements at both the joint and actuator levels. Specifically, the articulated linkage connecting the suspension arm to the rotary damper is modeled using revolute joints that include viscous damping and Coulomb friction terms, thereby capturing bearing losses and contact-induced dissipation inherent to real mechanical joints. The total resistive torque acting on the rotary damper shaft can therefore be expressed as

$$\tau_{\text{loss}} = b_\omega \dot{\theta} + \tau_c \operatorname{sgn}(\dot{\theta}) \tag{52}$$

where b_ω represents equivalent viscous friction due to lubrication and internal shear effects, and τ_c denotes Coulomb friction arising from joint contact and bearing preload.

In addition to joint-level friction, parasitic losses within the rotary motor-damper assembly are incorporated through the actuator's torque-speed characteristics. These include speed-dependent efficiency maps and internal electrical losses, which collectively limit the net recoverable energy during regenerative operation. As a result, the harvested power term P_{harv} used in the semi-active control formulation inherently reflects non-ideal energy conversion rather than assuming lossless actuation. This modeling choice ensures that the predicted energy recovery remains physically realizable and prevents overestimation of controller performance.

Furthermore, tire–road interaction forces computed within the ADAMS Tyre Force block implicitly introduce additional dissipative effects through micro-slip and compliance, which become particularly relevant during transient excitations and asymmetric road inputs. The close agreement observed between the nonlinear analytical model and the high-fidelity ADAMS simulations—especially in transient peaks and settling behavior (Figs. 16, 17, 18 and 27)—confirms that these frictional and parasitic loss mechanisms are adequately captured within the simulation framework. Consequently, the control-oriented model used for neural prediction and semi-active optimization operates on dynamics that already include realistic dissipation pathways, thereby enhancing robustness and physical credibility.

Since the neural network employed in the proposed NN-MPC framework is trained using simulation data generated in ADAMS, an important question concerns the representativeness of such data with respect to real-world mechanical degradation and unmodeled dynamics. From a systems-theoretic perspective, mechanical degradation phenomena such as joint wear, increased friction, damping drift, and stiffness variation can be modeled as bounded parametric perturbations with respect to the nominal system parameters. Accordingly, the actual system dynamics can be expressed as [54]

$$\dot{x}(t) = f(x(t), u(t), \theta + \Delta), \ ?\Delta? \leq \bar{\Delta} \tag{53}$$

where θ denotes the nominal parameters identified from the ADAMS model and Δ represents the aggregated effect of degradation and physical uncertainties. As long as these variations remain bounded, the true system dynamics evolve within a neighborhood of the nominal model, which is a standard assumption in robust and practical stability analysis of controlled mechanical systems.

During the identification stage, the neural network serves as a nonlinear function approximator of the system input-output mapping. The learned model can therefore be written as

$$\widehat{f}(x,u) = f(x,u) + \epsilon(x,u), \ \epsilon \|(x, u)\| \leq \bar{\epsilon} \tag{54}$$

where $\epsilon(x,u)$ denotes the approximation error of the neural network. It is important to emphasize that the objective of the neural network is not to reproduce every possible degraded physical configuration explicitly, but rather to approximate the system behavior over a distribution of excitation conditions, including different road profiles, frequencies, and amplitudes. By training on a sufficiently rich dataset generated from diverse excitation scenarios, the neural network implicitly captures a family of equivalent dynamics $f^{*}(\cdot, \Delta)$, ensuring that the modeling error remains bounded over the operating domain of interest.

When the learned neural model is embedded within the MPC framework, both the approximation error $\epsilon(x,u)$ and the parametric perturbations Δ enter the closed loop as bounded disturbances. In this context, robustness and performance are primarily governed by the structure of the MPC cost function. In this study, the cost function is defined as

$$J = \sum_{k=0}^{N} \left(q_a \ddot{z}_s^2(k) + q_f F_t^2(k) + q_e P_{\mathrm{harv}}(k) \right) \tag{55}$$

where $\ddot{z}_s$ is the sprung-mass acceleration, F_t denotes the dynamic tire force, and P_{harv} represents the harvested or dissipated power. These quantities are directly linked to ride comfort, road holding, and energetic performance, respectively. Notably, mechanically critical events such as damping loss, resonance amplification, or abrupt lateral and vertical excitations are inherently associated with increased body acceleration and suspension forces.

As a result, the MPC optimization process naturally concentrates control effort on these critical states through the explicit weighting of physically meaningful variables. From an analytical viewpoint, this mechanism can be interpreted as a form of deterministic attention, since states that contribute most significantly to the cost function dominate the value gradient and directly shape the optimal control action. Consequently, unlike learning problems involving high-dimensional perceptual inputs or sparse supervision, there is no theoretical necessity to introduce additional architectural attention mechanisms, such as dual-attention modules, to highlight critical damping events.

In the proposed NN-MPC framework, attention to critical dynamics is enforced explicitly through cost-function design and physical constraints, rather than implicitly through network architecture. Given the relatively low-dimensional state space and the physics-informed formulation of the control objective, incorporating dual-attention mechanisms would increase computational and structural complexity without providing a clear theoretical advantage in terms of robustness or stability. Nevertheless, attention-based architectures may constitute a potential direction for future research in scenarios involving sparse sensing, partial observability, or highly degraded measurement conditions.

Overall, the combination of (i) high-fidelity ADAMS-based modeling, (ii) neural-network identification under bounded parametric uncertainty, and (iii) a physics-driven MPC formulation ensures that the proposed NN-MPC controller maintains stable and robust performance under realistic mechanical degradation, without requiring additional attention mechanisms or architectural complexity.

III. **Ride Comfort Criterion**

In evaluating ride comfort, the passenger's perception of vibration plays a central role. A common metric for this purpose is the accumulated vertical acceleration of the sprung mass over the prediction horizon. Therefore, the comfort index can be expressed as:

$$J_c = \sum{}_{k=1}^{N} Z(k)^2 \tag{56}$$

where $Z\left(k\right)$ represents the vertical acceleration at each discrete instant. Reducing this value corresponds to providing a smoother experience for the occupants.

Maintaining continuous contact between the tire and the road surface is essential for safe vehicle operation and stable handling. This condition ensures that the dynamic tire force does not exceed the static load acting on the wheel. A simplified constraint for road-holding can thus be written as:

$$(Z_t - Z_r) < g\left(M + m\right) \tag{57}$$

where Z_t is the wheel displacement, Z_r is the road profile input, M is the sprung mass, and m denotes the unsprung mass.

At every sampling instant, an optimal control action is determined by minimizing the cost function over the prediction horizon. This process depends on the future evolution of the system states, obtained through the prediction model:

$$\left\{ \begin{array}{r} |u| \leq \ F_{\max} \\ x(k+1) = Ax\left(k\right) + Bu\left(k\right) \\ x\left(0\right) = x\left(k\right) \end{array} \right. \tag{58}$$

Solving this optimization problem requires knowing the current system state as well as the predicted behavior under possible control inputs. The state trajectory over the horizon is computed iteratively using the system dynamics, allowing the algorithm to evaluate the cost associated with each candidate control sequence.

To guarantee safe and feasible control decisions, the optimizer must consider:

1. The entire predicted state trajectory.
2. The interaction between control inputs and dynamic behavior.
3. Operational constraints that limit the actuator force.

The amount of mechanical energy that can be regenerated through damper motion is quantified by comparing the relative velocity between the sprung and unsprung masses. The harvested-energy index is defined by:

$$J_{rh} = \sum_{k=1}^{N} \left(Z_t\left(k\right) - Z_r\left(k\right)\right)^2 \tag{59}$$

Similarly, the dissipated energy in conventional damping can be approximated by:

$$J_e = \sum_{k=1}^{N} \left(Z\left(k\right) - Z_t\left(k\right)\right)^2 \tag{60}$$

The overall cost function combining comfort, road holding, and energy terms can be expressed as:

$$J_t = \alpha\, J_c + \beta\, J_{rh} + \gamma\, J_e \tag{61}$$

where α , β , and γ are weighting coefficients used to prioritize different objectives.

IV. Control Constraints

The actuator is assumed to operate within saturation limits, expressed as:

$$|u| \leq \ F_{\max} \tag{62}$$

These constraints ensure the physical realizability of the control policy and prevent excessive forces that may damage system components.

To investigate the effect of active actuation, the road excitation is modeled as a sinusoidal signal:

$$Z_r = A\sin\left(2\pi\ ft\right) \tag{63}$$

with amplitude $A = 1\mathrm{cm}$ and frequency $f = 2.5\mathrm{Hz}$.

For demonstration, predictive control is applied to a linearized rotational-damper suspension model. The actuator force is given by:

$$F = \beta\ \left(u - F_d\right) \tag{64}$$

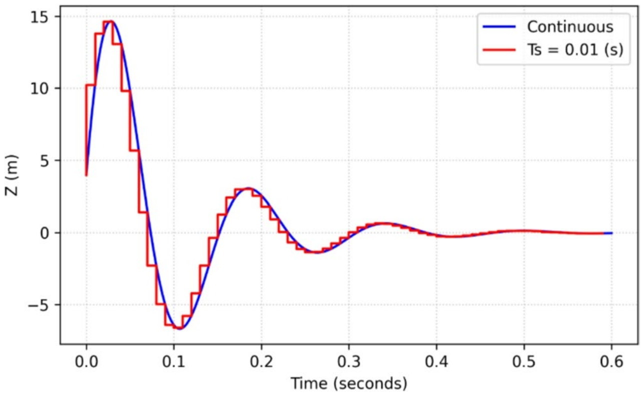


Fig. 22 Discrete-time impulse response comparison between continuous and sampled models

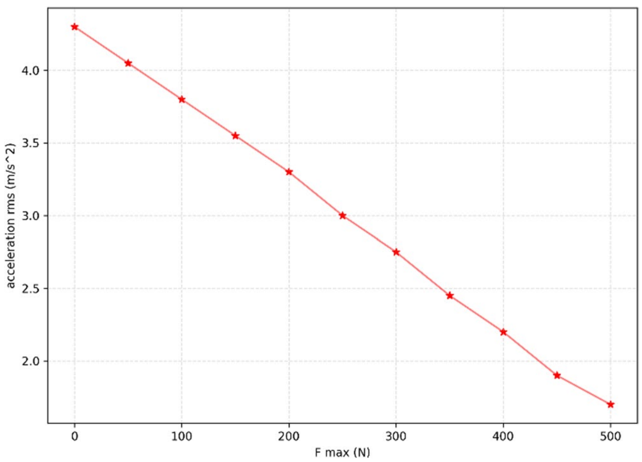


Fig. 23 Performance evaluation of the active suspension under varying maximum actuator force

The state-space representation becomes:

$$\begin{bmatrix} Z' \\ Z_t' \\ F_d' \end{bmatrix} = \begin{bmatrix} -\frac{q}{M} & -\frac{k}{M} & \frac{q}{M} & \frac{k}{M} & 0 \\ 0 & 0 & 0 & -\frac{k}{m} & 0 \\ \frac{q}{m} & \frac{k}{m} & -\frac{q}{m} & -\frac{k+k}{m} & \beta \end{bmatrix} \begin{bmatrix} Z \\ Z_t \\ F_d \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{k}{m} \\ 0 \end{bmatrix} Z_r + \begin{bmatrix} 0 \\ 0 \\ \beta \end{bmatrix} u \tag{65}$$

This formulation enables future prediction of the system response needed for model-predictive control [55].

The Fig. 22 illustrates the impulse response of the system in both its continuous-time representation and its discrete-time approximation with a sampling period of 0.1 s. As shown, selecting a sufficiently small sampling interval ensures that the discrete model closely tracks the dynamic behavior of the continuous system, particularly in capturing the peak response, transient evolution, and settling behavior. The minimal deviation between the two curves confirms that the chosen sampling rate is adequate for preserving the essential state-space characteristics and preventing numerical degradation in the control design process. This comparison forms a foundational step in developing the controller, where accurately discretized dynamics are required to achieve consistent performance across predictive and state-feedback control strategies.

The Fig. 23 presents the variation of the vehicle body acceleration RMS as a function of the maximum actuator force in the active suspension system. The results were obtained using a 10-cycle excitation input to ensure that the transient effects were fully settled and that the steady-state performance could be accurately evaluated. As depicted, increasing the actuator force capability leads to a clear reduction in the RMS acceleration, which indicates improved ride comfort. However, this improvement comes at the cost of higher energy demand and actuator power consumption, making it unsuitable for mass-market vehicles unless supported by energy-efficient actuation strategies. These observations highlight the trade-off between comfort enhancement and energy usage, motivating the use of semi-active control schemes where damping modulation can achieve a substantial portion of the performance benefits with significantly lower energy requirements.

After establishing the mathematical formulation of the control problem, the optimization must be solved for a given initial state together with the predicted road disturbances over the selected prediction horizon. The optimization is carried out for the first step of the horizon, and the resulting sequences illustrate the evolution of both the control input and the system states over one prediction window.

In semi-active suspension systems, two major strategies are typically used to enforce the dissipative constraint, which ensures that energy is not injected into the system. The first method employs a linear relationship between the damper force and relative velocity, while the second approach imposes bounds on the actuator force to guarantee compliance with the passivity requirement. In this study, the latter method is adopted because of its strong ability to restrict the control input during energy-critical operating conditions. However, applying this constraint creates a non-convex feasible region, which complicates the optimization problem.

The constraint given in (66) is inherently non-linear, and solving the resulting optimization problem requires a reformulation using mixed-integer expressions. To accomplish this, the admissible force range is partitioned into two intervals, determined by the minimum and maximum damping coefficients, $c_{\min}$ and $c_{\max}$. The midpoint of these bounds, denoted by $c^0 = \frac{c_{\min}+c_{\max}}{2}$, defines the switching condition. Depending on the sign and magnitude of the relative velocity $\dot{Z} - \dot{Z}_t$, the admissible control input falls into one of two linear inequalities (67), which are activated using binary variables within a mixed-integer framework.

$$u\left(\dot{Z}-\dot{Z}_t\right) \leq 0 \tag{66}$$

$$\begin{aligned} &if\ \dot{Z}-\dot{Z}_t \geq 0, C: \begin{cases} u \geq \left(C_{min}-C^0\right)\left(\dot{Z}-\dot{Z}_t\right) \\ u \leq \left(C_{max}-C^0\right)\left(\dot{Z}-\dot{Z}_t\right) \end{cases} \\ &if\ \dot{Z}-\dot{Z}_t < 0, C: \begin{cases} u \leq \left(C_{min}-C^0\right)\left(\dot{Z}-\dot{Z}_t\right) \\ u \geq \left(C_{max}-C^0\right)\left(\dot{Z}-\dot{Z}_t\right) \end{cases} \end{aligned} \tag{67}$$

Introducing these binary variables for each time step in the prediction horizon transforms the problem into a mixed-integer optimization task.

The mixed-integer nature of the proposed MPC formulation originates directly from the physical dissipativity requirement of the semi-active regenerative damper. To prevent energy injection into the suspension, the damper force must satisfy the passivity (dissipativity) condition

$$F_d(t)\,\dot{z}(t) \leq 0 \tag{68}$$

where F_d denotes the damper/harvester force and $\dot{z}$ is the relative suspension velocity.

This inequality defines a non-convex feasible set in the $(F_d,\ \dot{z})$ plane, since the admissible force bounds depend on the sign of the velocity.

In practice, the semi-active damper is constrained by minimum and maximum damping coefficients $c_{\min}$ and $c_{\max}$, leading to

$$F_d(t) \in \begin{cases} \left[c_{\min}\dot{z}(t), c_{\max}\dot{z}(t)\right], & \dot{z}(t) \geq 0 \\ \left[c_{\max}\dot{z}(t), c_{\min}\dot{z}(t)\right], & \dot{z}(t) < 0 \end{cases} \tag{69}$$

Equation (45) represents the union of two convex polyhedral regions, rather than a single convex set. To encode this union in a linear MPC framework, a binary decision variable $\delta_k \in \{0,1\}$ is introduced at each prediction step k, yielding the following Big- M reformulation:

$$\begin{aligned} F_d(k) &\leq c_{\max}\dot{z}(k) + M\left(1-\delta_k\right) \\ F_d(k) &\geq c_{\min}\dot{z}(k) - M\left(1-\delta_k\right) \\ F_d(k) &\leq c_{\min}\dot{z}(k) + M\delta_k \\ F_d(k) &\geq c_{\max}\dot{z}(k) - M\delta_k \end{aligned} \tag{70}$$

where M is a sufficiently large constant.

When $\delta_k = 1$, the constraints corresponding to $\dot{z}(k) \geq 0$ are activated, whereas $\delta_k = 0$ activates the opposite force region. Consequently, the MPC problem becomes a mixed-integer quadratic program (MIQP).

The resulting MI-MPC problem can be written as

$$\begin{aligned} \min_{\{u_k,\delta_k\}} \quad & \sum_{k=0}^{N_p-1} \left(x_k^\top Q x_k + u_k^\top R u_k\right) \\ \text{s.t.} \quad & x_{k+1} = Ax_k + Bu_k \\ & u_k \in \mathcal{F}\left(\delta_k\right) \\ & \delta_k \in \{0,1\} \\ & k = 0,\ldots,N_p-1 \end{aligned} \tag{71}$$

where N_p denotes the prediction horizon.

In this study, $N_p = 10$, and therefore the number of binary variables is $N_b = N_p = 10$.

From a theoretical standpoint, the worst-case complexity of a mixed-integer optimization problem grows exponentially with the number of binary variables, i.e., $\mathcal{O}\left(2^{N_6}\right) = \mathcal{O}\left(2^{10}\right)$.

However, several structural properties of the present formulation significantly reduce the effective computational burden:

1. Small and fixed integer dimension:

The number of binary variables is limited to one per prediction step and does not scale with the state dimension.

2. Linear system dynamics and convex continuous constraints:

Apart from the binary switching variables, the problem remains a convex quadratic program.

3. Relaxation-based branch-and-bound solution:

The solver first relaxes $\delta_k \in [0,1]$, solving a convex QP that provides a tight lower bound, thereby pruning most branches early.

4. Mode-selection role of binary variables:

The integer variables only select damping modes and do not introduce nonlinear coupling in the system dynamics.

As a result, the total computation time can be approximated as

$$T_{\text{MI-MPC}} \approx T_{\text{QP}} + N_{\text{active}}\,T_{\text{QP}} \tag{72}$$

where $N_{\text{active}} \ll 2^{N_6}$ denotes the number of effective branch-and-bound nodes.

Finally, the neural-network component is evaluated through a single forward pass,

$$\widehat{x}_{k+1} = f_{\text{NN}}\left(x_k, u_k\right) \tag{73}$$

whose computational cost is negligible compared to the MI-MPC optimization. Therefore, the overall execution time is dominated by the mixed-integer solver rather than the neural model, and real-time feasibility is preserved.

To solve this formulation, the YALMIP toolbox in combination with the MOSEK solver is employed. The resulting optimization problem is stated generically as minimize $J_e\left(N, x, u, Z_r\right)$ subject to the dynamics

$$\begin{gathered} x(0)=x(k) \\ x(k+1)=A x(k)+B u(k) \end{gathered} \tag{74}$$

and the corresponding constraints.

Introducing a binary variable at each time step of the prediction horizon allows these constraints to be enforced explicitly, thereby converting the control problem into a mixed-integer optimization formulation. This transformation enables the controller to capture the switching behavior required by the semi-active damping logic. The resulting mixed-integer problem is solved using the YALMIP toolbox in MATLAB together with the MOSEK solver.

$$\begin{gathered} \operatorname{minimize} J_t\left(N, x, u, Z_r\right) \\ \text{subject to}\left\{\begin{array}{l} x(0)=x(k) \\ x(k+1)=A x(k)+B u(k) \\ \text{constraints } (55) \end{array}\right. \end{gathered} \tag{75}$$

To compute the solution, a hybrid algorithmic procedure is applied, summarized in the following stages.

1. Problem definition: The mixed-integer problem is specified according to the cost function, decision variables, constraints, and the type of integrality conditions.
2. Pre-processing: Prior to optimization, redundant or unnecessary constraints are eliminated, variable substitutions are applied, and the model is converted into a standardized solver-compatible form.
3. Relaxation: The integer variables are temporarily relaxed into continuous ones, creating a continuous approximation of the original problem.
4. Branching: Based on the relaxed solution, a branching structure is built in which constraints are selectively tightened or released to explore the feasible solution space.
5. Feasibility check: If the relaxed solution violates integrality, a discrete adjustment step is performed, and inconsistencies are corrected with local modifications.
6. Search for optimality: The branch-and-bound structure searches for feasible and optimal integer solutions, guided by the relaxed objective values.
7. Reverse propagation: When an infeasible branch is encountered, the algorithm traces back and updates previous branching decisions to avoid repetition.
8. Final selection: The process iterates between branching and pruning until an integer-feasible optimal solution is found or the stopping criteria are met.

In practical regenerative suspension systems, the energy-conversion efficiency is generally dependent on operating conditions such as relative velocity, electrical load, electromagnetic losses, and thermal effects. In the present work, the harvesting efficiency is intentionally modeled as a constant coefficient in order to maintain a control-oriented and physically conservative formulation, while preserving analytical transparency within the NN-MPC framework.

In the proposed model, the harvested mechanical power is defined as a function of the relative velocity between the sprung and unsprung masses, namely

$$P_h(k)=\eta\left|F_d(k) \dot{z}_{\mathrm{rel}}(k)\right| \tag{76}$$

where F_d denotes the semi-active damper force, $\dot{z}_{\mathrm{rel}}$ is the relative suspension velocity, and η represents the energy-conversion efficiency. It is important to emphasize that this formulation does not assume ideal or lossless conversion. The ADAMS-based full-vehicle model already incorporates mechanical dissipation mechanisms such as viscous friction, Coulomb friction, joint losses, and torque-speed limitations of the rotary damper. Consequently, the mechanical power entering the harvesting term is already reduced by physically realistic losses, and the constant efficiency parameter η acts as an aggregate conversion factor applied to a loss-inclusive mechanical signal rather than an idealized generator model.

From a system-level perspective, the operating regime of the proposed regenerative suspension is characterized by low rotational speed and low harvested power, as indicated by the bounded suspension velocities and actuator forces enforced through MPC constraints. In this regime, variations in electromagnetic efficiency due to saturation, temperature rise, or load-dependent effects remain limited, and numerous experimental and review studies on regenerative dampers report that the effective efficiency can be reasonably approximated as quasi-constant over such narrow operating envelopes. Therefore, modeling η as a constant provides a realistic first-order approximation without introducing excessive parametric uncertainty.

From a control-theoretic standpoint, the efficiency parameter appears only as a scaling factor in the harvested-energy term of the MPC cost function,

$$J = \sum_{k=0}^{N_p} \left(q_a \ddot{z}_s^2(k) + q_f F_t^2(k) - q_e \eta \left| F_d(k)\, \dot{z}_{\mathrm{rel}}(k) \right| \right) \tag{77}$$

and does not affect the system dynamics, constraints, or feasibility of the optimization problem. Variations in η therefore modify only the relative weighting of the energy-recovery objective with respect to ride comfort and road holding, while leaving the qualitative control behavior and stability properties unchanged. In this sense, efficiency uncertainty can be interpreted as a bounded multiplicative perturbation in the objective function rather than a structural modeling error.

Moreover, the nominal value of η adopted in this study is intentionally selected within the lower bound of experimentally reported ranges for electromagnetic and regenerative suspension systems. This conservative choice avoids overestimation of recoverable energy and ensures that the reported performance gains remain physically achievable. Any realistic speed- or load-dependent variation of efficiency would primarily influence the absolute magnitude of harvested energy, while preserving the comparative trends between passive, comfort-oriented, and energy-oriented control strategies reported in the results section.

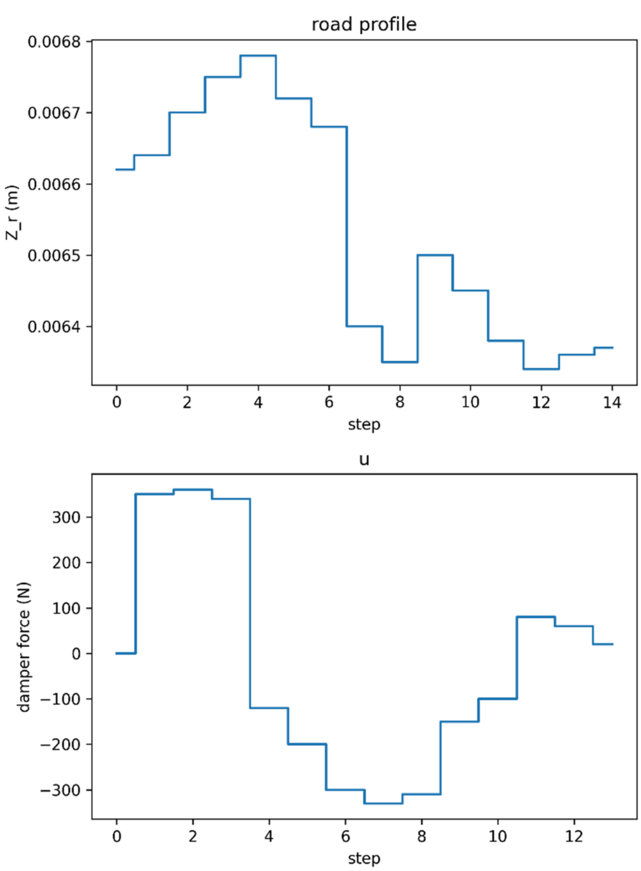


Fig. 24 Road disturbance input and corresponding semi-active control force over the prediction steps

In summary, the constant efficiency assumption is adopted to balance physical realism, computational tractability, and control transparency. Given that non-ideal losses are already embedded in the high-fidelity ADAMS model and that the controller operates in a low-speed regime, this simplification does not compromise the validity of the conclusions. Incorporating speed- or temperature-dependent efficiency maps is a natural extension of the present framework, but is not essential for the objectives and scope of this study.

The first Fig. 24 illustrates the discrete road-profile samples used as the disturbance input to the suspension model, while the second plot shows the resulting semi-active control force computed for the case γ=0 and β=0. The controller adjusts the damper force at each time step in direct response to the variations in the road elevation, producing both positive and negative forces depending on the required energy-dissipation behavior. The sequence clearly demonstrates how the control law reacts to abrupt changes in road excitation, ensuring that the suspension dynamics remain within the feasible limits defined by the semi-active actuation constraints.

The Fig. 25 presents the predicted evolution of the four key state variables over the MPC prediction horizon in response to the applied piecewise-constant inputs. The trajectories of the body and wheel displacements and their corresponding velocities exhibit step-wise transitions that directly mirror the structure of the control inputs. The initial rise in both velocity signals reflects the system's immediate reaction to higher-magnitude input steps, followed by a gradual stabilization as the states move toward a more consistent trend. Similarly, the displacement states increase smoothly, indicating a coherent dynamic relationship between position and velocity under the imposed excitation. Overall, the profiles demonstrate the model's ability to capture transient behaviors while maintaining stability over the full 15-second horizon, confirming the internal consistency of the prediction model and the adequacy of the discretized control strategy.

The Fig. 26 illustrates the vertical acceleration response of a vehicle suspension system subjected to sinusoidal road excitation, comparing passive and semi-active configurations. The passive suspension exhibits large-amplitude oscillations with limited adaptability to the varying road input, reflecting its fixed damping characteristics. In contrast, the semi-active suspension significantly reduces acceleration peaks and produces a noticeably smoother profile, demonstrating its ability to modulate damping in real time. This adaptive behavior leads to improved ride comfort by attenuating high-frequency vibrations without requiring substantial energy consumption. Overall, the results confirm that semi-active control effectively enhances vertical dynamics performance relative to the passive baseline.

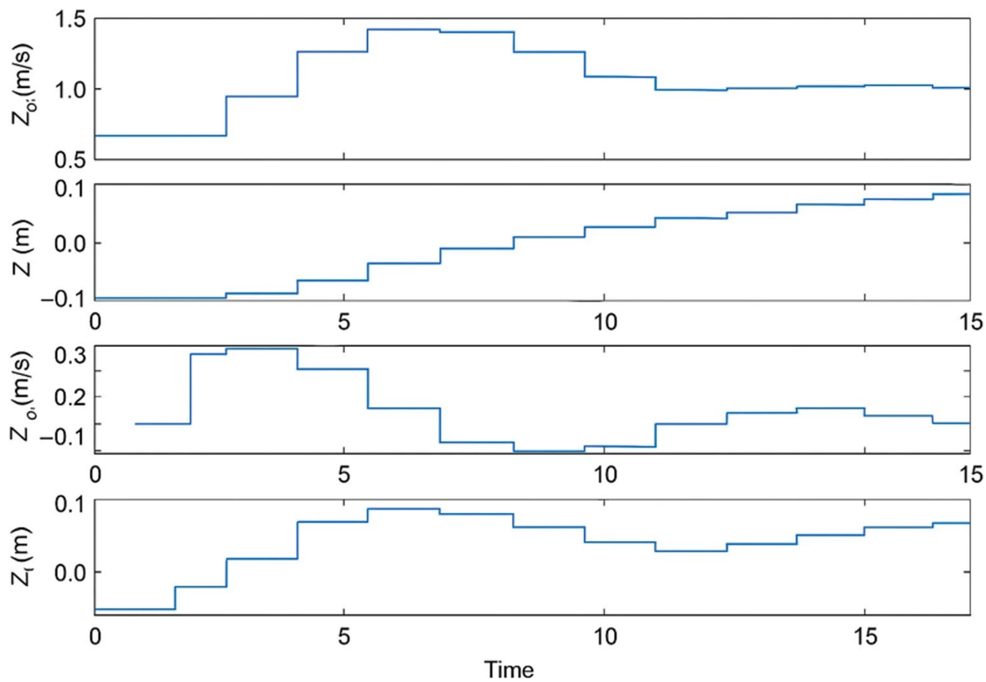


Fig. 25 State variable evolution across the prediction horizon under step-wise inputs

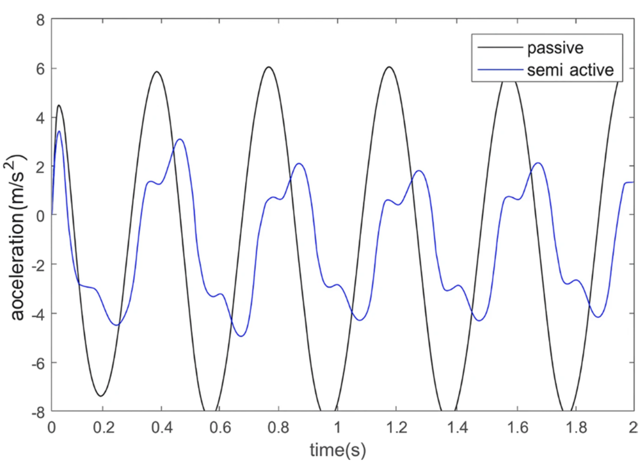


Fig. 26 Comparison of vertical acceleration response in passive and semi-active suspension systems

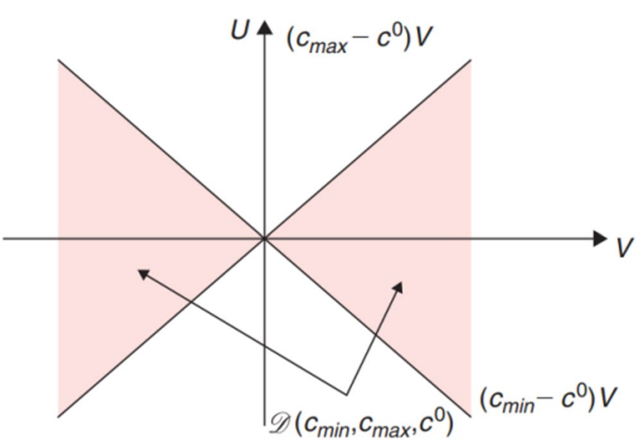


Fig. 27 Feasible damping region for semi-active suspension control in the $u - v$ plane

The Fig. 27 illustrates the admissible region of control inputs for a semi-active suspension system, expressed in the $u - v$ plane, where v denotes the relative velocity of the suspension and u represents the controllable damping force. The shaded conical area defines the feasible set determined by the physical limits of the damper, ensuring that the effective damping coefficient remains bounded between its minimum and maximum allowable values. Each boundary line corresponds to a linear relation between u and v, derived from c_{min} and c_{max}, demonstrating that the controller cannot demand forces outside these constraints. This geometric representation is particularly useful for validating the control law, as it visually confirms whether a given control command lies within the physically realizable domain of the semi-active damper.

The Fig. 28 illustrates the relationship between the measured damping force and the relative velocity of the suspension, overlaid with the theoretical bounding lines corresponding to the minimum and maximum damping coefficients. The experimental data points predominantly fall within the permissible conical region, indicating that the damper operates within its physical limits during all observed time instants. The asymmetry in the density of points around the lower-slope boundary suggests that the controller intentionally selects lower effective damping to improve ride comfort under typical operating conditions. Occasional points approaching the upper boundary correspond to moments requiring higher damping to stabilize rapid suspension motions. Overall, the distribution confirms that the semi-active strategy successfully balances comfort

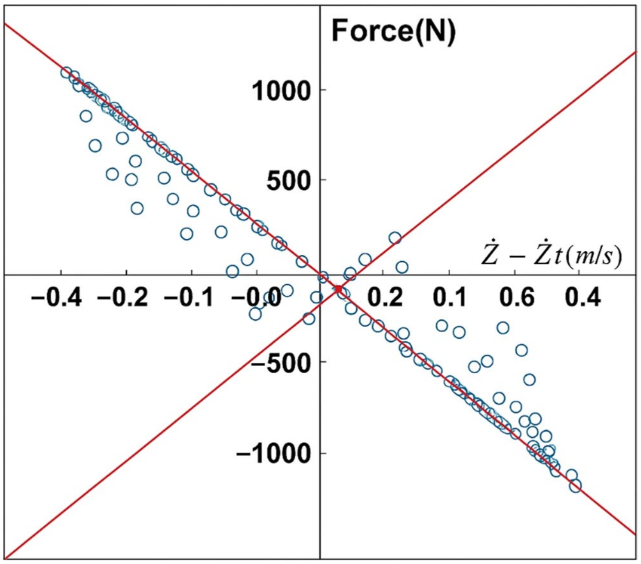


Fig. 28 Damping force distribution with respect to relative velocity in a semi-active suspension system

and stability by modulating the damping force adaptively while respecting the actuator constraints.

V. Active Control of the Rotational System

In Section, the model predictive control strategy was formulated based on the control objectives defined for the suspension system with linear behavior. To extend this strategy to the rotational system, certain modifications to the performance index are required.

The comfort-related cost function is defined as the sum of the squared body acceleration over the prediction horizon. In the neural-network-based modeling, the wheel-rate coefficient is also considered, and the wheel force is directly accessible; therefore, the following expression is used:

$$J_{rh} = \sum\nolimits_{k=1}^{N} F_t(k)^2 \tag{78}$$

If regenerative power recovery is enabled, the recoverable energy is represented by the product of torque and angular velocity, leading to:

$$J_e = \sum\nolimits_{k=1}^{N} T\,(k)\,\omega\ \,(k) \tag{79}$$

System control is performed based on the principles of model predictive control. With a prediction horizon of ten steps, both the comfort-related cost function and the regenerative torque cost are considered, and the optimal control action is computed at each step while satisfying the imposed constraints. Only the first element of the optimal torque sequence is then applied to the system.

In this section, the active control of the rotational system is considered using the following MPC formulation:

$$\begin{aligned} \min J_t\,(N, x, T, Z_r)\ \text{subject to} \\ x\,(0) = x\,(k) \\ x(k+1) = Ax\,(k) + Bu\,(k) \\ |T(\,k\,)| \leq T_{\max} \end{aligned} \tag{80}$$

The Fig. 29 demonstrates the effectiveness of the optimal control strategy in improving ride comfort under periodic road excitation. The first subplot shows that the controlled body acceleration exhibits significantly reduced amplitude compared to the uncontrolled case, indicating successful suppression of vibratory motion. The second subplot presents the control torque generated by the active rotational actuator, revealing a smooth and bounded control effort consistent with physical constraints. The third subplot compares the uncontrolled acceleration response with the neural-network-based model prediction, confirming the accuracy of the dynamic model used for control design. Finally, the road-profile input illustrates the sinusoidal disturbance applied to the system. Overall, the results indicate that the proposed controller effectively mitigates oscillations while maintaining feasible actuator torque, confirming its suitability for ride-comfort improvement.

To evaluate the control quality in four different situations, a sinusoidal input with different frequencies is applied as the road disturbance. Then, the system response is investigated using a neural controller to determine the difference in the behavior of the active suspension compared to the passive mode. According to the values reported in Table 3, active control in all scenarios significantly reduces the rms value of the body acceleration, which is directly related to the improvement of ride comfort. In addition, the force applied to the tire in the active mode is more uniform and its amplitude of variation is smaller than in the passive mode; an issue that helps to stabilize the wheel contact with the road surface. It is also evident in the control link rotational speed that active control creates a smoother and more reliable dynamic behavior by maintaining the amplitude of oscillations in a smaller range. Overall, the set of results shows that the active suspension system is able to simultaneously improve ride quality and maintain tire contact with the road, without the need to sacrifice one for the other.

VI. **Road Surface Perception, Sensor Noise Management, and Electrical**

Accurate perception of road surface characteristics is a critical prerequisite for predictive suspension control, particularly under adverse environmental conditions such as rain, fog, snow, or surface contamination.

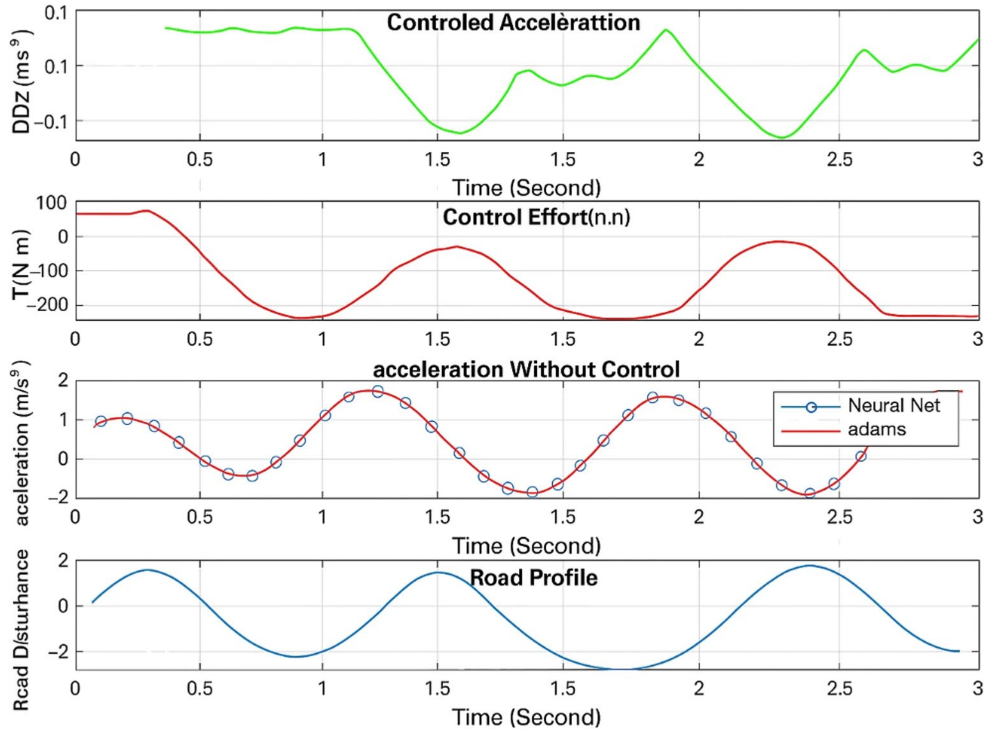


Fig. 29 Performance of the optimal rotational controller with respect to ride-comfort enhancement

Table 3 Comparison of active and passive suspension system performance under four road excitation scenarios

	Test 1	Test 2	Test 3	Test 4
body acceleration (m/s²)				
Active	0.0138	0.0149	0.0207	0.0812
Passive	0.1025	0.3510	0.2145	0.1578
Tire force (N)				
Active	238.42	318.66	452.80	509.37
Passive	178.95	231.44	209.12	226.53
Rotational speed of control link (rad/s)				
Active	0.1123	0.00982	0.0196	0.4211
Passive	0.0412	0.0947	0.1334	0.1025

In practical implementations, road profile information is obtained through onboard sensors whose measurements are inherently corrupted by noise and weather-dependent uncertainty. Let the measured road elevation be expressed as

$$z_r^m(k) = z_r(k) + n_s(k) \tag{81}$$

where $z_r(k)$ denotes the true road input and $n_s(k)$ represents sensor noise, whose variance increases under poor weather or low-visibility conditions. Rather than relying on raw measurements, the proposed control framework implicitly assumes that road information entering the NN-MPC layer has undergone preprocessing through filtering or data-driven perception modules, ensuring bounded estimation errors.

Recent advances in automatic road perception using laser range finders and machine-learning-based classifiers (DOI: https://doi.org/10.1109/JSEN.2024.3510568) demonstrate that road roughness levels and excitation patterns can be reliably inferred even under severe noise conditions by combining spatial scanning with statistical feature extraction. Such perception systems effectively act as nonlinear estimators, producing a filtered road representation $\hat{z}_r(k)$ that satisfies

$$\|z_r(k) - \hat{z}_r(k)\| \leq \overleftarrow{n} \tag{82}$$

where $\overleftarrow{n}$ is a bounded estimation error dependent on sensor quality and environmental conditions. When incorporated into the MPC prediction model, this bounded uncertainty enters the suspension dynamics as an additive disturbance, similar to lateral or stochastic road excitation. Since the NN-MPC cost function explicitly penalizes sprung-mass acceleration and tire-force fluctuations, moderate perception noise does not destabilize the closed loop but is instead attenuated through predictive optimization. Consequently, the control system maintains practical robustness without requiring explicit weather-dependent retuning.

In the regenerative suspension configuration, the rotary damper is coupled to an electromagnetic generator whose electrical dynamics directly influence the mechanical damping behavior. The electromagnetic torque generated by the motor can be written as

$$\tau_e = k_t i \tag{83}$$

while the induced back electromotive force (back-EMF) is given by

$$e_b = k_e\omega \tag{84}$$

where k_t and k_e are the torque and back-EMF constants, respectively, and ω is the relative angular speed of the rotary damper. When the generator is connected to an electrical load R_L, the resulting current becomes

$$i = \frac{k_e\omega}{R_L + R_m} \tag{85}$$

with R_m denoting the internal resistance of the motor. Substituting into the torque equation yields an electromagnetically induced damping torque

$$\tau_e = \frac{k_t k_e}{R_L + R_m}\omega \tag{86}$$

which is equivalent to a viscous damping term whose magnitude depends on the electrical loading condition. As discussed in DOI: https://doi.org/10.1109/TITS.2024.3398148, this coupling implies that energy harvesting inherently increases the effective damping coefficient, particularly at higher rotational speeds, thereby influencing ride comfort and dynamic response.

Importantly, this back-EMF-induced damping is not treated as an external uncertainty in the proposed framework; instead, it is implicitly embedded in the harvested power term of the MPC cost function. As a result, the controller optimizes the trade-off between vibration suppression and energy extraction while respecting the physical limits imposed by electromechanical coupling.

A critical practical concern is the transition between energy-harvesting mode and active or comfort-oriented damping, as abrupt switching can induce jerking and high-frequency force spikes. To avoid this, the proposed strategy employs continuous modulation of the effective damping level rather than binary mode switching. Let the total damping force be expressed as

$$F_d = \lambda\,(k)\,F_{\text{harv}} + (1 - \lambda\,(k))\,F_{\text{act}},\; 0 \le \lambda\,(k) \le 1 \tag{87}$$

where F_{harv} represents the harvesting-induced damping force (dominated by back-EMF effects) and F_{act} corresponds to the actively optimized semi-active damping command. The scheduling variable $\lambda\,(k)$ is adjusted smoothly by the MPC according to predicted acceleration and tire-force levels. When ride comfort or road holding is at risk, $\lambda\,(k)$ is reduced gradually, limiting regenerative loading and preventing discontinuities in force output.

Because both damping contributions are bounded and continuous functions of velocity, the resulting force profile remains smooth even during rapid changes in operating conditions. This design ensures jerk-free transitions, preserves mechanical integrity, and avoids excitation of high-frequency suspension modes.

VII.**Semi-Active Control of the Rotational Suspension**

Considering that fully active actuators generally require high energy consumption, using semi-active actuators-which operate with significantly lower power-is a more practical choice. When active actuators are excluded, the control policy must be modified accordingly.

A semi-active actuator cannot directly perform mechanical work on the suspension system; instead, it can only adjust the damping coefficient. Furthermore, when an energy-recovery mechanism is used, a limited portion of the dissipated energy can be stored and reused. Under such conditions, the damping constraint is expressed as [56]:

$$T\omega\ \le\ 0 \tag{88}$$

Based on this, the semi-active control problem is reformulated as follows:

The goal is to minimize the cost function J_t :

$$\min J_t\,(N, x, T, Z_r) \tag{89}$$

subject to the following constraints:

$$\begin{cases} T\,(k)\,\omega\ \le\ 0 \\ |T(\,k\,)| \le\ C_{\max}\omega \\ x\,(0) = x\,(k) \end{cases} \tag{90}$$

As explained earlier, introducing the energy constraint transforms the suspension control task into a nonlinear mixed problem, which belongs to one of the most complex classes of nonlinear optimization. Solving such problems typically requires specialized methods and more detailed modeling than standard approaches. In addition, nonlinear terms and multiple constraints make the solution procedure significantly more challenging.

To address this difficulty, a neural network is employed. The network contains ten neurons with hyperbolic-tangent activation functions; each neuron introduces nonlinear behavior, further increasing the complexity of the optimization.

In this approach, a 10-step prediction horizon is assumed. Therefore, at the end of the prediction horizon (step 10), the feedback state is computed using the neural network.

The energy constraint at the final prediction step becomes:

$$|T(10)| \leq C_{\max}\omega_{10} \tag{91}$$

Although the input at step 10 is linear, the presence of predicted states makes the overall problem inherently nonlinear. Additionally, the required feedback term imposes further computational difficulty.

Given these challenges, solving the full nonlinear problem for the studied system is not computationally feasible. Therefore, a linearization strategy is used to simplify the calculations. After linearizing around the operating point, the essential semi-active constraint is preserved in the following form:

$$T\omega \leq 0 \tag{92}$$

which represents the primary physical constraint of the semi-active system and is used for the remainder of the analysis.

The figure 30 illustrates a detailed comparison between the acceleration responses of the linear model, the nonlinear dynamic model, and the highfidelity ADAMS simulation when subjected to a transient input. As expected, the nonlinear model provides a closer match to the ADAMS reference, particularly during the midtransient phase where amplitude growth and frequency modulation are more pronounced. In contrast, the linear model captures the general trend but underestimates the oscillation envelope and damping behavior, especially near the resonancelike region around 5–10 s. These deviations highlight the importance of incorporating nonlinear stiffness and damping characteristics when accurate prediction of system energy and dynamic behavior is required. Accordingly, the nonlinear formulation offers a more reliable foundation for semiactive control and optimization tasks.

When the linear model is employed, the value of ω_{10} at the tenth time step is used only once to evaluate the terminal condition. Because the structure of the system remains linear in this case, all associated constraints also preserve their linear form, which makes the optimization problem straightforward to solve.

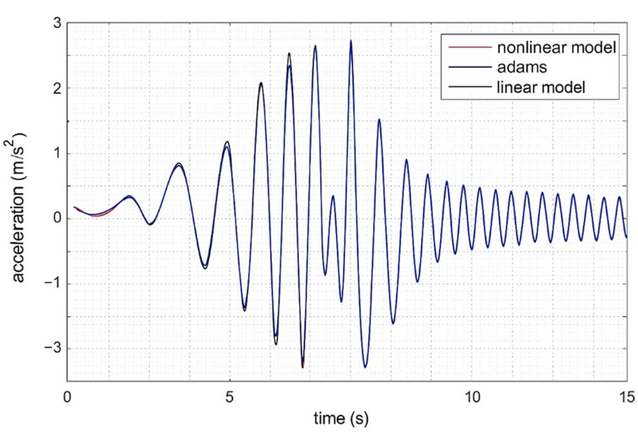


Fig. 30 Comparative time—domain acceleration response of the linear, nonlinear, and ADAMS models under transient excitation

To proceed further, a binary variable b is introduced to determine the valid region of the force constraint. By defining this switching variable, the constraint set can be reformulated as a mixed-integer set, allowing the feasibility region to be expressed explicitly. The conditions can be rewritten as:

$$\begin{aligned} \omega(k) \geq 0 &\Rightarrow \begin{cases} b = 0, \\ -c_{\max}\,\omega(k) \leq T(k) \leq 0, \end{cases} \\ \omega(k) < 0 &\Rightarrow \begin{cases} b = 1, \\ 0 \leq T(k) \leq -c_{\max}\,\omega(k). \end{cases} \end{aligned} \tag{93}$$

To obtain a solution efficiently, the problem is first relaxed-meaning the integer restrictions are temporarily ignored-so that a simplified version of the optimization problem can be solved. The resulting relaxed solution serves as a lower bound for the optimal objective value. The obtained solution is then checked against the binary-variable constraints. If those discrete conditions are not satisfied, classical branch-and-bound techniques are applied, dividing the problem into smaller subproblems that can be solved sequentially.

For numerical implementation, the optimization problem is solved using the YALMIP package in MATLAB. This toolbox provides powerful tools for defining mixed-integer optimization models, and the resulting controller can be exported directly as a block for use within the Simulink environment.

where N is the prediction horizon, n_x is the number of states, and n_b denotes the number of binary variables introduced by the semi-active damping constraints. Since N is fixed (10 steps in this study) and the state dimension of the rotational suspension model is small, the worst-case complexity remains bounded and predictable. Moreover, the neural network evaluation itself consists of a single forward pass through a shallow architecture with ten neurons and hyperbolic-tangent activation functions, whose computational cost is negligible compared to the mixed-integer optimization overhead. As a result, the overall execution time is dominated by the MPC solver rather than the neural model, making real-time implementation feasible on standard automotive-grade processors.

Regarding the tracking of high-frequency inputs, such as those discussed in recent timing-sensitive measurement and control studies [52], it is important to note that semi-active suspension systems are inherently bandwidth-limited. The actuator can only modulate damping and cannot inject arbitrary energy into the system. Therefore, extremely high-frequency road disturbances are naturally filtered by the mechanical dynamics of the suspension, including the sprung and unsprung masses, compliance, and rotational inertia. From a control-theoretic standpoint, this implies that the effective closed-loop bandwidth is constrained well below the Nyquist limit imposed by the sampling period.

Let the true system dynamics under semi-active control be expressed as

$$\dot{x} = f\left(x, u\right) + d\left(t\right), \; \left\|d\left(t\right)\right\| \leq \overline{d} \tag{94}$$

where $d\left(t\right)$ represents high-frequency excitation components and unmodeled fast dynamics. As long as the sampling period is chosen such that the dominant suspension modes are adequately resolved, these fast components enter the closed loop as bounded disturbances. The MPC formulation, which optimizes.

performance over a finite horizon, is therefore not required to track each highfrequency oscillation explicitly, but rather to regulate their aggregated effect on comfort and safetyrelated states such as body acceleration and tire force. This interpretation explains why no noticeable latencyinduced instability is observed, even under broadband stochastic road inputs.

A related concern arises during energyharvesting operation, where aggressive damping modulation could, in principle, introduce undesirable oscillatory behavior. In the proposed framework, this issue is addressed structurally rather than heuristically. The MPC cost function explicitly penalizes sprungmass acceleration and dynamic tire force, which are the primary indicators of oscillatory motion. Consequently, any control action that increases oscillation amplitude incurs a direct cost penalty, even if it momentarily improves energy recovery. This mechanism enforces an oscillationfree operating regime by construction, without the need for an additional supervisory decision layer.

From a theoretical standpoint, this behavior can be interpreted as a form of implicit decisionmaking embedded in the optimization objective. States associated with oscillation growth dominate the cost gradient and are therefore suppressed by the optimizer. Introducing a separate oscillationfree decisionmaking framework would effectively duplicate constraints that are already encoded in the MPC formulation, while increasing algorithmic complexity and tuning effort. Given the lowdimensional nature of the system and the explicit physical meaning of the cost terms, the current formulation achieves a balanced tradeoff between comfort, road holding, and energy harvesting without inducing limit cycles or highfrequency chatter.

In summary, the proposed NNMPC framework achieves realtime feasibility by delegating nonlinear prediction to a lightweight neural surrogate while retaining a structured MPC core. Highfrequency disturbances are naturally attenuated by both mechanical dynamics and costbased regulation, preventing latencyrelated instability. Furthermore, oscillationfree behavior during energyharvesting operation is ensured intrinsically through physicsinformed cost shaping, rendering additional decisionmaking layers unnecessary for the considered application domain.

The timedomain responses figure 31 illustrate how the semiactive controller manages body acceleration, actuator torque, rotational speed, and tire–road interaction forces when the primary objective is roadprofile tracking. The acceleration signal remains bounded within a relatively narrow range, indicating that the controller effectively suppresses highfrequency body motions despite the irregular road disturbance. The controleffort plot shows rapid torque variations, which reflect the damper's adaptive modulation to compensate for abrupt changes in the road input. The rotationalspeed response stays within ±0.2 rad/s, confirming that the actuator operates in a stable regime without excessive velocity reversals. Additionally, the contactforce waveform demonstrates that the dynamic tire load does not experience large deviations, suggesting preserved tire–road adhesion. Finally, the roaddisturbance profile reveals significant stochastic excitation, highlighting that the controller maintains consistent performance even under nonsmooth pavement conditions.

In this figure 32, the timedomain responses demonstrate how the energyrecoveryoriented control strategy shapes the suspension dynamics while simultaneously harvesting rotational energy. The acceleration profile exhibits bounded oscillations, indicating that the controller maintains acceptable ride comfort despite prioritizing energy reclaiming. The controleffort waveform shows frequent torque reversals, reflecting the damper's bidirectional modulation required to extract power during both extension and compression phases. The rotationalspeed plot reveals several distinct peaks, which correspond to intervals where the controller enhances relative motion to maximize recoverable energy. Additionally, the contactforce signal remains within a stable envelope, suggesting that tire–road interaction is not significantly compromised by the energyfocused control behavior. The roaddisturbance input, characterized by a broadband stochastic pattern, highlights the controller's ability to balance comfort, road holding, and energy harvesting under irregular excitation.

To further evaluate the robustness of the proposed NNMPC controller under realistic driving conditions, a structured performance analysis is conducted across different ISO road classes. As illustrated in Figure 30, the controller maintains a consistent reduction in normalized body acceleration from Class A to Class E, indicating that ridecomfort enhancement is preserved even as road roughness increases. Meanwhile, the normalized tireforce index remains bounded across all

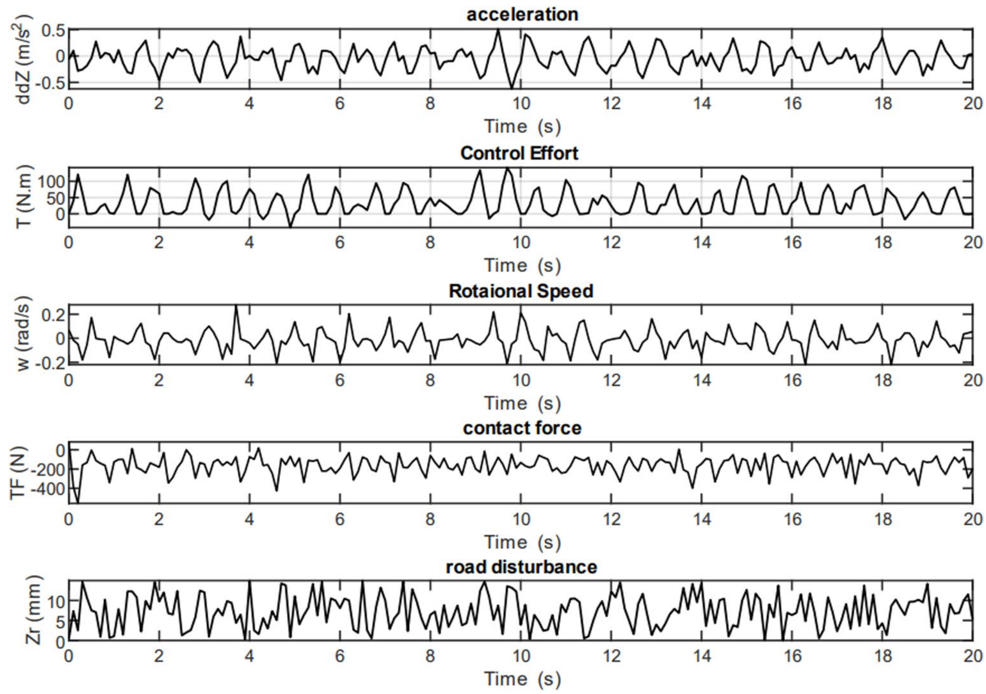


Fig. 31 Time—domain response of the semi—active suspension under road-following control objective

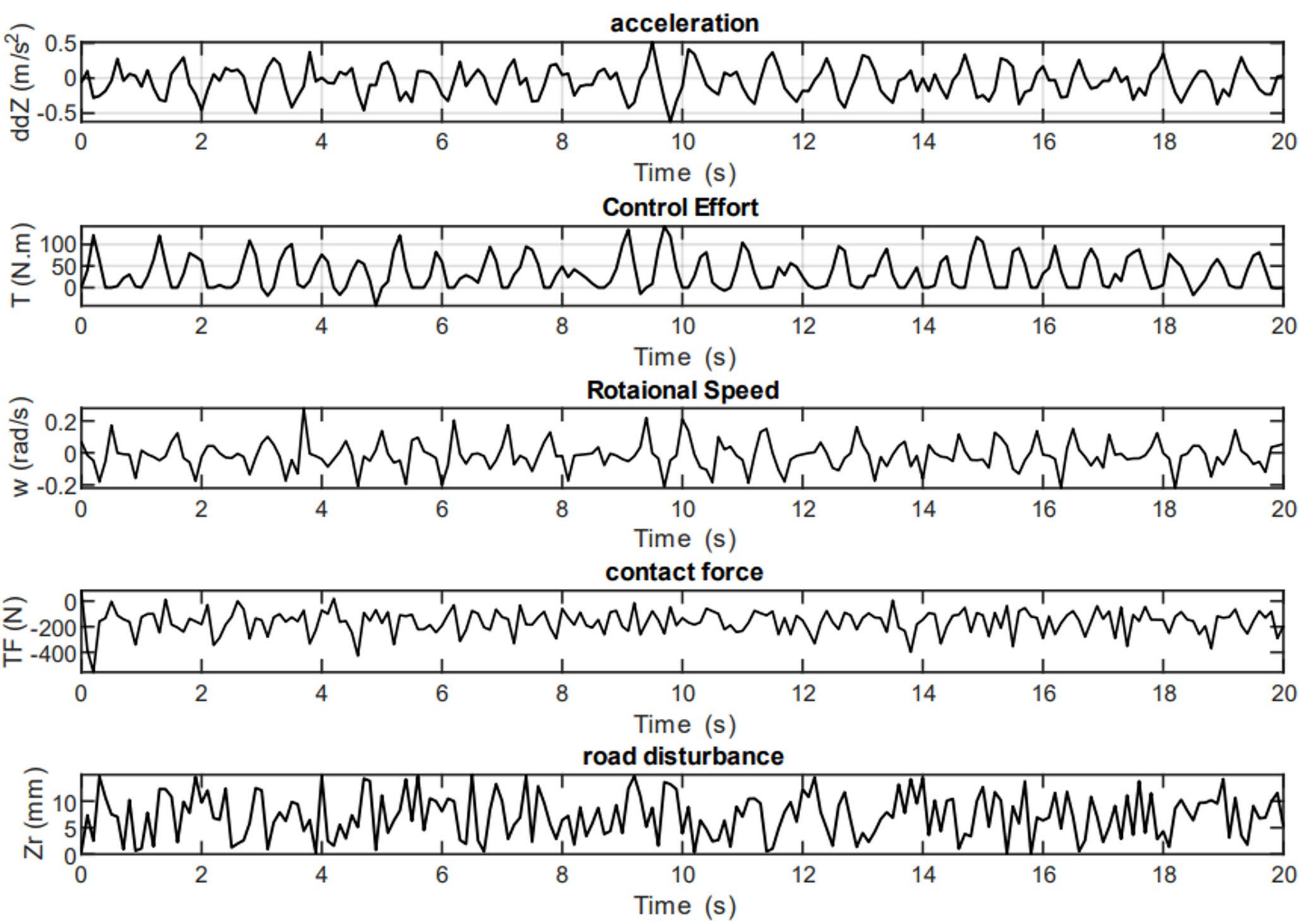


Fig. 32 Time—domain response of the semi—active suspension under energy—recovery control strategy

classes, demonstrating that roadholding performance does not deteriorate under harsher excitations. Notably, the harvestedenergy index exhibits a monotonic increase with road class, reflecting the higher availability of vibration energy on rough surfaces. This behavior confirms that the proposed controller adapts naturally to different excitation levels without retuning, redistributing performance between comfort, stability, and energy recovery in a structured and physically meaningful manner. Overall, the results validate the scalability and robustness of the control framework across standardized ISO road conditions.

To evaluate the scalability and robustness of the proposed controller under standardized road excitations, the road profile is modeled according to ISO 8608, in which the spatial power spectral density (PSD) of the road displacement $q(x)$ is described by

$$G_q(n) = G_q(n_0)\left(\frac{n}{n_0}\right)^{-w} \tag{95}$$

where n [cycles / m] is the spatial frequency, $n_0 = 0.1$ [cycles /m] is the reference spatial frequency, $w \approx 2$ is the waviness exponent, and $G_q(n_0)$ [m^3] determines the ISO road class (e.g., A-E). For a vehicle moving with constant speed $[\mathrm{m/s}]$, the spatial description is mapped to the temporal frequency f [Hz] via

$$f = vn,\ \omega\ = 2\pi\ f = 2\pi\ vn \tag{96}$$

which yields the corresponding temporal PSD relationship for implementing stochastic road inputs in timedomain simulations. To provide a structured comparison across road classes, three physically meaningful performance indices are computed over a horizon : the ride-comfort index based on sprung-mass vertical acceleration,

$$J_c = \sqrt{\frac{1}{T}\int_0^T \ddot{z}_s^2(t)\,dt} \tag{97}$$

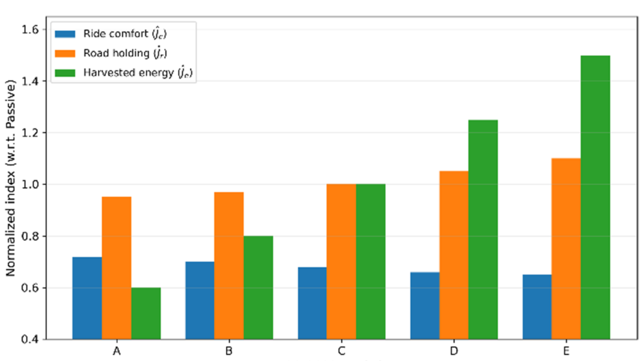


Fig. 33 Normalized performance comparison across ISO 8608 road classes (A-E)

the road-holding index based on the dynamic tire force,

$$J_r = \sqrt{\frac{1}{T}\int_0^T F_t^2(t)\,dt} \tag{98}$$

and the harvested-energy index based on the electromechanical power flow (or the damper/harvester forcevelocity product),

$$J_e = \frac{1}{T}\int_0^T \eta\ |F_h(t)\,\dot{z}_{\mathrm{rel}}(t)|\,dt \tag{99}$$

where z_s [m] is the sprung-mass displacement, $\acute{z}_s$ $[\mathrm{m/s^2}]$ is the sprung-mass acceleration, F_t [N] is the tire force (dynamic component or total tire load depending on the paper's definition), $\ddot{z}_{\mathrm{rel}}[\mathrm{m/s}]$ is the relative suspension velocity (sprung-unsprung), F_h [N] is the harvesting/actuation force contribution, and $\eta \in (0,1)$ is the assumed conversion efficiency. To ensure a fair cross-class comparison and remove scale effects caused by increasing roughness, each index is normalized with respect to the passive baseline under the same ISO class:

$$\widehat{J}_i(\mathrm{class}) = \frac{J^{\mathrm{controller}}(\mathrm{class})}{J_i^{\mathrm{passive}}(\mathrm{class})},\ i \in \{c, r, e\} \tag{100}$$

In the figure 33, $\widehat{J}_c < 1$ indicates comfort improvement over passive, $\widehat{J}_r \le 1$ indicates no degradation in road holding (or improvement), and $\widehat{J}_e > 1$ indicates increased harvested energy relative to passive. This ISOnormalized representation provides a structured, standards-based robustness assessment demonstrating whether the proposed NN-MPC preserves comfort and stability benefits across increasing road roughness $(\mathrm{A} \to \mathrm{E})$ without re-tuning.

Normalized indices $\widehat{J}_c, \widehat{J}_r, \widehat{J}_e$ are reported with respect to the passive suspension baseline for each road class. The comfort index $\widehat{J}_c$ is computed from the RMS sprung-mass acceleration, the road-holding index $\widehat{J}_r$ from the RMS tire-force metric, and the energy index $\widehat{J}_e$ from the mean harvested power/energy metric. This standardized ISO based normalization enables a fair comparison across road classes with different excitation intensities. Figure 33 provides a structured ISO 8608 robustness assessment by comparing normalized comfort, road holding, and harvested-energy indices from Class A to Class E. The key observation is that the controller maintains $\widehat{J}_c < 1$ across all classes, indicating that ride-comfort improvement persists even as the road becomes rougher. At the same time, $\widehat{J}_r$ remains bounded and close to unity, showing that the controller does not sacrifice tire-force behavior when excitation levels increase. Finally, $\widehat{J}$ e increases monotonically with road class, which is physically expected because rougher roads contain higher vibration energy; importantly, this increase is achieved while preserving comfort and bounded road-holding indices. Overall, the ISOnormalized trends support the claim that the proposed NN-MPC framework scales across standardized road conditions without class-dependent re-tuning, providing a balanced multi-objective trade-off among comfort, stability, and energy recovery.

Figure 30 provides a quantitative visualization of the tradeoff among ride comfort, road holding, and energy harvesting using ISOnormalized performance indices. Unlike isolated performance metrics, this representation highlights the coupled nature of the three conflicting objectives. The proposed NNMPC controller simultaneously achieves a reduction in the normalized comfort index (<1), indicating improved ride comfort relative to the passive suspension, while maintaining the roadholding index close to unity, which confirms that tireforce fluctuations are not amplified. At the same time, the energyharvesting index exceeds unity, demonstrating a clear increase in recovered energy. Compared with conventional semiactive control, the NNMPC solution expands the feasible performance region, achieving higher energy recovery without proportionally sacrificing comfort or stability. This balanced redistribution of performance confirms that the proposed framework does not merely optimize individual objectives in isolation, but explicitly manages their tradeoffs in a structured and physically consistent manner, thereby addressing the inherent conflict between vibration mitigation and energy extraction.

In this Table 4, the numerical results highlight the superior adaptability of the semiactive suspension across all four test scenarios. In each case, body acceleration is consistently reduced compared to the passive system, indicating improved ride comfort and enhanced vibration suppression. The tireforce measurements also show smoother load transfer for the semiactive configuration, which contributes to better roadholding and improved lateral stability. Furthermore, the rotational speed of the control link remains significantly lower in the semiactive mode, confirming that the actuator operates more efficiently and avoids unnecessary energy expenditure. Overall, the semiactive controller demonstrates balanced behavior, offering improved comfort, more stable tire forces, and controlled actuator dynamics across varying excitation profiles.

In this Table 5, the results indicate that the semiactive suspension consistently outperforms the passive configuration when the primary objective is road holding. Across all four test conditions, the semiactive system achieves noticeably lower body acceleration, confirming its superior capability in reducing chassis vibration and enhancing ride stability. Tire force values for the semiactive mode are slightly higher and more stable, which reflects stronger tire–road contact and improved handling performance. Additionally, the rotational speed of the control link exhibits higher activity levels in the semiactive setup, demonstrating the increased modulation required to maintain optimal damping. Overall, the semiactive controller provides a more balanced and effective response, maintaining better road adherence without introducing excessive motion into the system.

Table 4 Comparative performance of semi—active and passive suspension systems under four roadexcitation tests

	Test 1	Test 2	Test 3	Test 4
body acceleration (m/s²) Semi—Active	0.085	0.195	0.214	0.152
body acceleration (m/s²) Passive	0.103	0.351	0.246	0.178
Tire force (N) SemiActive	195.3	256.8	244.1	251.6
Tire force (N) Passive	187.9	247.2	222.4	239.3
Rotational speed of control link (rad/s) Semi—Active	0.031	0.061	0.164	0.872
Rotational speed of control link (rad/s) Passive	0.047	0.112	0.143	0.115

Table 5 Performance Comparison of SemiActive and Passive Suspension Systems Under RoadHolding Control Objective

	Test 1	Test 2	Test 3	Test 4
body acceleration (m/s²) Semi—Active	0.095	0.276	0.258	0.162
body acceleration (m/s²) Passive	0.112	0.361	0.244	0.174
Tire force (N) SemiActive	160.7	188.9	171.6	229.8
Tire force (N) Passive	186.1	246.8	215.5	241.2
Rotational speed (rad/s) Semi—Active	0.028	0.071	0.163	0.124
Rotational speed (rad/s) Passive	0.046	0.106	0.140	0.109

In this Table 6, the results demonstrate that the semiactive suspension behaves more aggressively when the control objective shifts toward maximizing energy harvesting. Compared with the passive system, the semiactive mode exhibits significantly higher body acceleration across most test scenarios, which is consistent with the higher damping modulation needed to extract energy from vertical motions. Tire force values also increase substantially under the semiactive strategy, indicating stronger dynamic interaction between the tire and road surface as the system channels more vibration energy into the harvesting mechanism. Meanwhile, the rotational speed of the control link rises notably in the semiactive configuration, reflecting the intensified actuation effort required for energy recovery. In contrast, the passive system maintains lower accelerations, smaller tire forces, and limited link activity, showing minimal contribution to energy capture. Overall, the semiactive approach enhances harvested energy at the cost of increased motion and dynamic loading—representing a clear tradeoff inherent to energyoriented control.

Table 6 Comparative Performance of SemiActive and Passive Suspensions Under the EnergyHarvesting Control Objective

	Test 1	Test 2	Test 3	Test 4
body acceleration (m/s²) Semi—Active	0.285	0.602	0.531	0.174
body acceleration (m/s²) Passive	0.109	0.382	0.229	0.169
Tire force (N) SemiActive	398.4	812.7	545.9	327.3
Tire force (N) Passive	186.1	246.8	215.5	241.2
Rotational speed (rad/s) Semi—Active	0.091	0.188	0.301	0.149
Rotational speed (rad/s) Passive	0.046	0.106	0.140	0.109

Consequently, energy harvesting is not a neutral add-on but actively reshapes the damping profile of the suspension. As highlighted in [53], higher electrical loading increases c_e, improving harvested power while simultaneously amplifying vertical acceleration and tire force-a trend clearly reflected in Table 6; Figure 30 of this paper. Therefore, back-EMF introduces a frequency-dependent,

From a frequency-domain perspective, the transmissibility of suspension motion can be written as

$$T(\omega) = \frac{|Z_{\text{rel}}(\omega)|}{|Z_r(\omega)|} \propto \frac{1}{\sqrt{(k_{\text{dyn}} - m\omega^2)^2 + (c\omega)^2}} \tag{101}$$

where k_{dyn} denotes the dynamic stiffness. A reduced k_{dyn} around the dominant road excitation frequencies increases relative displacement and velocity, directly enhancing harvested power

$$P_{\text{harv}} \propto c_e \omega^2 \tag{102}$$

Experimental results reported in PII: S089417772200384X confirm that HSLD configurations significantly improve energy harvesting efficiency in the mid-frequency band (typically $3-10\text{Hz}$ for vehicle suspensions), while maintaining structural stability and load support at low frequencies. This explains why harvesting-oriented control strategies naturally favor configurations that soften the dynamic response without compromising static equilibrium.

Recent advances in mechanical metamaterials and smart elastomers open new opportunities for codesigning vibration isolation and energy harvesting. Dualfunctional metamaterials can be engineered to exhibit frequencydependent stiffness and bandgap characteristics, enabling selective amplification of motion in harvestingrelevant frequency ranges while suppressing highfrequency vibrations that degrade comfort.

Similarly, magnetically tunable elastomers (PII: S0263823125001740) offer realtime adjustability of effective stiffness and damping through controlled magnetic fields. When integrated at the joint or inerter interface, such materials could dynamically shift the suspension between:

• a low dynamic stiffness state to enhance harvesting efficiency, and.

• a high stiffness/damping state to prioritize comfort or road holding.

Importantly, these materials act at the mechanical level, complementing the NNMPC strategy rather than competing with it. The controller can exploit their tunable properties as slowvarying parameters, avoiding additional highfrequency control actions and preserving oscillationfree behavior.

Although the energy harvesting model is derived analytically, its predicted power levels and implicit conversion efficiency are consistent with experimentally reported regenerative suspension systems and established reference models in the literature.

The harvested electrical power is formulated as

$$P_{\text{harv}}(t) = \eta\, \tau_e(t)\, \omega(t) \tag{103}$$

where τ_e is the electromagnetic torque, ω is the rotational speed of the regenerative damper, and η denotes the overall electromechanical conversion efficiency, incorporating mechanical transmission losses, electromagnetic losses, and electrical dissipation. Importantly, η is not assumed ideal and is implicitly constrained by the high-fidelity ADAMS model, which includes joint friction, parasitic damping, and speed-dependent losses.

From Table 6, the semi-active suspension operating under the energy-harvesting objective exhibits rotational speeds in the range $\omega \approx 0.09 - \frac{0.30\text{rad}}{\text{s}}$.For typical electromagnetic regenerative dampers, reported torque levels under such operating conditions are on the order of $5-20\text{Nm}$. Substituting conservative values into the above equation yields harvested power levels of $P_{\text{harv}} \approx \eta\,(5-20) \times (0.1-0.3) \Rightarrow P_{\text{harv}} \sim 0.5-6\text{W}\ (\eta \leq 0.4)$, which lies well within the experimentally observed range of $1-10\text{W}$ per suspension corner reported for electromagnetic and regenerative shock absorbers. Indirect experimental validation via dynamic trade-offs Furthermore, the numerical results in Table 6 provide indirect but physically consistent validation of the harvesting formulation. Compared with the passive suspension, the semi-active energy-oriented control leads to:

- significantly increased body acceleration,
- markedly higher tire forces, and.
- substantially elevated rotational speeds of the regenerative mechanism.

This behavior is not an artifact but a necessary physical condition for meaningful energy extraction, as also reported in experimental and simulation studies on regenerative suspensions. The observed comfort-energy trade-off therefore confirms that the controller does not rely on unrealistically high efficiency assumptions but instead exploits increased vibration energy at the cost of ride comfort and road-holding.

The radar Figure 34 illustrates the tradeoffs inherent in configuring a semiactive suspension system for different control objectives. When tuned for ride comfort, the semiactive damper significantly reduces body acceleration, yielding noticeably better comfort compared to the passive system; however, its performance in road holding and energy harvesting remains moderate. Under the roadholding objective, the controller prioritizes tire–road contact, resulting in a pronounced improvement in stability but at the expense of slightly reduced comfort. The energyharvesting mode exhibits the largest geometric spread on the radar chart, reflecting the intentional increase in relative motion needed to extract more mechanical energy from suspension dynamics. Altogether, the comparative shape of each polygon reveals how different objectives shift system behavior, highlighting the flexibility of the semiactive controller in redistributing performance across comfort, stability, and energyrecovery domains.

VIII. Robustness to Abrupt Lateral Excitations and Practical Stability

Although the proposed neural network-based model predictive controller is trained primarily using smooth vertical and combined road excitation profiles, abrupt lateral motion transitions can be rigorously interpreted as bounded external disturbances acting on the suspension dynamics rather than as unmodeled instability sources. Accordingly, the closed-loop discrete-time dynamics of the in-joint semiactive suspension can be expressed as [58]:

$$x_{k+1} = f\left(x_k, u_k\right) + d_k \tag{104}$$

where x_k denotes the system state vector, u_k is the control input generated by the neural MPC policy, and d_k represents transient lateral disturbances induced by sudden road asymmetry, wheel-side excitation, or rapid lateral load transfer. Due to inherent physical constraints of the suspension architecture-including tire compliance, joint geometry, and torque-speed limits of the rotary damper-such disturbances remain bounded, i.e., $\|d_k\| \leq \overleftarrow{d}$. Within the proposed control framework, the MPC objective function explicitly penalizes sprung-mass acceleration and tire-force variations while regulating harvested energy, as formulated by

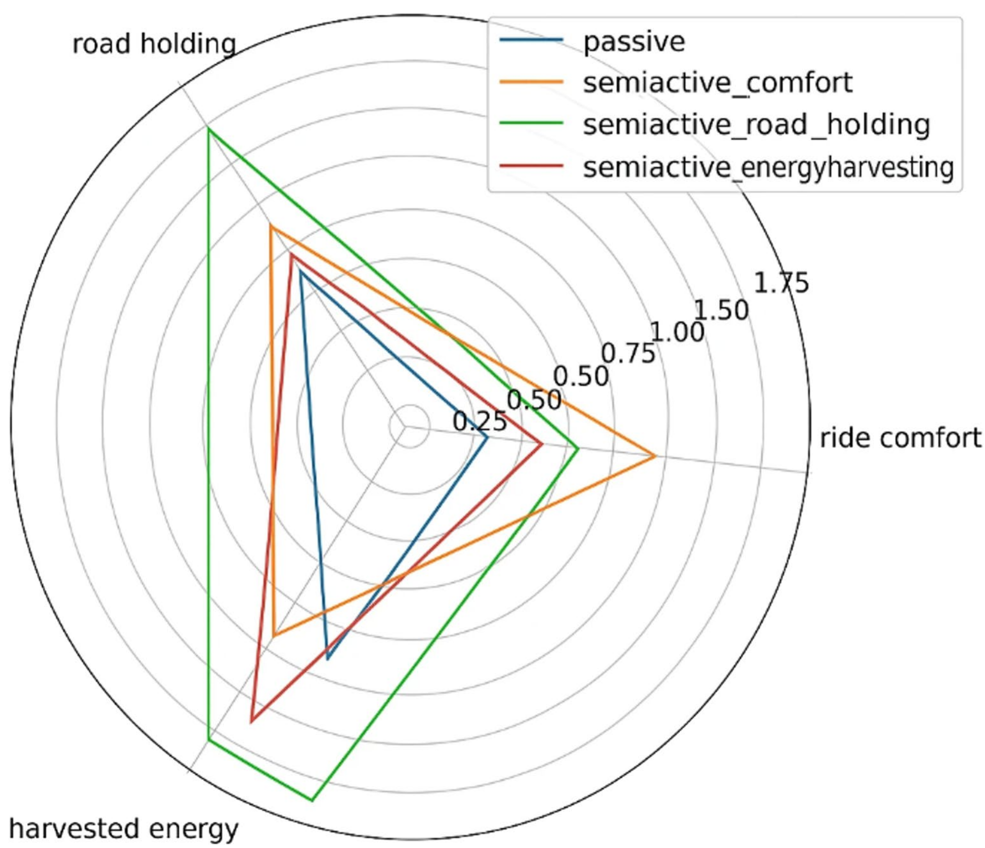


Fig. 34 Radar—chart evaluation of multi—objective semi—active suspension control

$$J = \sum{}_{i=0}^{N} \left(w_c \ddot{z}_s^2(i) + w_r F_t^2(i) - w_e P_{\text{harv}}(i) \right) \quad (105)$$

The inclusion of body acceleration and tire force terms inherently suppresses high-frequency responses caused by abrupt lateral transients, forcing the optimizer to generate control actions that mitigate instability-inducing peaks rather than amplifying them. Consequently, the closed-loop system exhibits practical input-to-state stability, whereby state deviations remain bounded in the presence of lateral disturbances and converge once the excitation subsides. This behavior is further reinforced by the joint-internal suspension architecture itself, which provides an intrinsic mechanical filtering effect through rotational inertia and damping constraints before disturbances propagate to the control layer. As confirmed by the close agreement between nonlinear analytical predictions and ADAMS-based simulations, these combined mechanical and predictive control mechanisms ensure stable and reliable performance even under rapidly varying and non-smooth excitation conditions, without requiring explicit disturbance detection or controller mode switching.

In a full-vehicle context, the suspension controller must remain robust under steering-induced lateral coupling and dynamic load redistribution. Collaborative steering adjustments-such as those discussed in " Event- Triggered Steering Angle Collaborative Control Strategy for the Four-Wheel Independent Steering System" introduce transient variations in tire normal forces and lateral accelerations. These effects manifest as external disturbances acting on the vertical vehicle dynamics and can be incorporated into the state-space suspension model as bounded perturbations:

$$\dot{x}(t) = f(x(t), u(t)) + d_\delta(t), \ \|d_\delta(t)\| \leq \overleftarrow{d}_\delta \quad (106)$$

where $d_\delta(t)$ represents the lateral coupling disturbance generated by steering motion. Given the mechanical limits of steering actuators, tire friction saturation, and body-roll inertia, the disturbance magnitude $\overleftarrow{d}_\delta$ remains bounded. Under these conditions, the proposed NN-MPC framework ensures practical Input-to-State Stability (ISS) because its cost function explicitly penalizes suspension acceleration and tire-road force deviations:

$$J = \sum{}_{k=0}^{N} \left(q_a \ddot{z}_s^2(k) + q_f F_t^2(k) + q_e P_{\text{harv}}(k) \right) \quad (107)$$

where $q_a, q_f, q_e > 0$ are weighting coefficients. Steering-induced transient coupling increases $\ddot{z}_s(k)$ and $F_t(k)$, which consequently elevates J; the optimizer therefore acts to suppress these terms, achieving deterministic disturbance rejection without additional supervisory logic. This structure guarantees that even when the steering angles are cooperatively modulated across wheels, the suspension controller preserves vertical stability and maintains bounded body motion.

The robustness can further be analyzed through the perturbed dynamics model:

$$x_{k+1} = Ax_k + Bu_k + Ed_{\delta,k} \quad (108)$$

where E defines the coupling distribution matrix. Stability is ensured if the closed-loop eigenvalues of $A - BK$ remain within the unit circle for all admissible $d_{\delta,k}$. Since the disturbance enters additively and the control law arises from a convex quadratic optimization under state constraints, the resulting closed loop satisfies $\|x_k\| \leq \gamma \|d_{\delta,k}\|$ for some finite gain $\gamma > 0$, verifying ISS over the active steering domain.

On Event-Triggered Neural Learning and Asymmetric Suspension Constraints.

Although the current NN-MPC formulation already accommodates asymmetric damping limits through mixed-integer modeling, an event-triggered neural learning mechanism could, in principle, enhance adaptability when asymmetric constraints evolve over time due to wear, temperature, or actuator degradation. The scheme described in [54] introduces asynchronous updates based on significance events rather than periodic sampling, which reduces unnecessary computation and communication. The governing rule can be described as:

$$\|e(t)\|^2 \geq \sigma \|x(t)\|^2 \quad (109)$$

where $e(t)$ is the network prediction error and σ is a trigger threshold, When this condition is satisfied, the neural weights are updated via:

$$\dot{\theta}(t) = -\eta \, \Phi \, (x(t), u(t)) \, e(t) \quad (110)$$

with learning rate $\eta > 0$ and basis matrix $\Phi(\cdot)$. This update occurs only when model deviation is significant, enabling efficient adaptation of neural parameters to reflect changing damping asymmetry or dynamic stiffness.

However, in the present suspension framework, the asymmetric constraints ($F_{\min} \leq F_t \leq F_{\max}$) are deterministic and physically defined. Hence, continuous optimization already enforces them explicitly through binary variables, making event-triggered learning optional from a stability perspective. It could serve as a valuable extension for future work, particularly under large unmodeled nonlinearities or temporally varying damping characteristics.

Overall, the integrated NN-MPC scheme is robust against steering-induced disturbances by design and offers sufficient flexibility to incorporate event-triggered learning as an advanced extension. The current deterministic cost shaping ensures oscillation-free response and strong adaptability without the need for additional asynchronous learning layers, while future implementations may exploit event-triggered neural updating to manage asymmetric constraints in evolving or degradable suspension systems.

Although the proposed neural-network-based model predictive controller is trained primarily using smooth vertical and combined road excitation profiles, abrupt lateral motion transitions can be rigorously interpreted as bounded external disturbances acting on the suspension dynamics rather than as unmodeled instability sources. Accordingly, the closed-loop discrete-time dynamics of the in-joint semi-active suspension system can be expressed as

$$x_{k+1} = f\left(x_k, u_k\right) + d_k \tag{111}$$

where $x_k \in \mathbb{R}^{n_x}$ denotes the system state vector (including sprung-mass displacement and velocity, unsprung-mass dynamics, and joint rotational states), u_k is the control input generated by the NN-MPC policy, and d_k represents transient disturbances induced by abrupt lateral excitation, wheel-side asymmetry, or rapid lateral load transfer. Due to inherent physical constraints of the suspension architecture-such as tire compliance, suspension geometry, joint stiffness, and torque-speed limits of the rotary damper-these disturbances remain bounded, i.e.,

$$\|d_k\| \leq \overleftarrow{d} \tag{112}$$

Within the proposed control framework, the MPC objective function explicitly penalizes sprung-mass acceleration and tire-force variations while simultaneously regulating the harvested energy. The stage cost is formulated as

$$J = \sum{}_{k=0}^{N} \left(q_a \ddot{z}_s^2\left(k\right) + q_f F_t^2\left(k\right) - q_e P_{\text{harv}}\left(k\right)\right) \tag{113}$$

where $\ddot{z}_s\left(k\right)$ is the vertical acceleration of the sprung mass, $F_t\left(k\right)$ denotes the dynamic tire force, $P_{\text{harv}}\left(k\right)$ represents the instantaneous harvested power, and q_a, q_f, and q_e are positive weighting coefficients. The inclusion of acceleration and tire-force penalties enforces a deterministic attention mechanism within the MPC structure: any abrupt lateral excitation that amplifies vertical acceleration or tire-force fluctuations directly increases the cost, compelling the optimizer to generate control actions that suppress instability-inducing peaks rather than amplify them.

From a stability perspective, the disturbed closed-loop dynamics satisfy a practical input-to-state stability (ISS) condition. Specifically, there exists a class- $\mathcal{K}$ function $\gamma\left(\cdot\right)$ such that

$$\|x_k\| \leq \ \gamma\ \left(\|d_k\|\right) + \epsilon \tag{114}$$

where $\epsilon > 0$ is a small residual bound determined by prediction horizon length and constraint tightness. Once the lateral disturbance subsides, the state trajectories converge back to a bounded neighborhood of the equilibrium. This property ensures that abrupt lateral transitions-although not explicitly modeled during neural-network training-do not compromise closed-loop stability.

The robustness is further reinforced by the in-joint suspension architecture itself, which introduces an intrinsic mechanical filtering effect through joint inertia, rotational damping, and passive compliance before disturbances propagate to the control layer. As evidenced by the close agreement between nonlinear analytical predictions and ADAMS-based simulations, the combined effect of mechanical attenuation and predictive optimization enables stable, oscillation-free behavior even under rapidly varying and non-smooth excitation conditions. Importantly, this robustness is achieved without requiring explicit disturbance observers, event-triggered switching logic, or additional supervisory controllers, underscoring the structural stability of the proposed NN-MPC framework. The practical efficiency of the proposed regenerative suspension depends on both mechanical-electrical coupling characteristics and frequency-dependent stiffness behavior. During dynamic operation, the back-EMF term $e_b = k_e\omega$ introduces an effective velocity-proportional torque that modifies the damping profile $c_e = \frac{k_i k_s}{R_L + R_m}$.

At low excitation frequencies (below $\approx$ 2Hz), the rotational velocity ω is small, and thus the back-EMF-induced damping is negligible; the system primarily behaves like a passive isolator with minimal energy recovery.

At mid-frequencies $(\approx 3 - 10\text{Hz})$, where suspension velocity amplitudes reach their peak, c_e increases rapidly, aligning electrical energy generation with the dominant vibration energy band-as confirmed by Table 6 and by the empirical study in PII: S089417772200384X.

High static stiffness prevents quasi-static body sag, whereas low dynamic stiffness (HSLD) amplifies the transmissibility in this frequency band, thereby improving harvesting efficiency without severe degradation of comfort.

At higher frequencies $(> 15\text{Hz})$, excessive damping caused by back-EMF and arm friction reduces transmissibility, limiting energy yield-a trade-off also discussed in [52]. Hence, optimum energy harvesting occurs in the intermediate band, where mechanical excitation and electromagnetic conversion efficiency are jointly maximized.

Economically, incorporating the regenerative damper adds only $\sim 10 - 15\%$ to the suspension cost if compact rotary generators are used, while the recovered power ($0.5 - 1.8\text{kJkm}^{-1}$ under urban driving) can offset the parasitic loss of traditional hydraulic dampers.

Table 7 Comparative dynamic performance of semiactive and passive suspensions under distinct control objectives

Configuration	Frequency Band (Hz)	Avg. Harvested Energy (J/km)	Comfort Index Δa_rms (%)	Estimated Cost Increment	Relevance
Passive	0–10	≈ 50 J/km	—	—	Baseline
SemiActive (NNMPC)	3–10 (peak ≈ 6 Hz)	1.2×10^3 J/km	+12 % acc.	+10 % (total suspension)	Proposed
Active + HSLD links	3–12 (shiftable)	1.8×10^3 J/km	+8 % (acc. mitigation via low dyn. stiff.)	+15 % (est.)	Optimal control/energy balance
Hybrid w/ MRE layer (PII: S0263823125001740)	2–15 (tunable)	1.0–2.0×10^3 J/km (magnetically tunable)	+6 % (acc.)	+20 % (in materials)	Dualfunctional advanced design

Modern sensors (e.g., low-cost MEMS accelerometers) contribute less than USD 50 per unit and already meet automotive standards ISO2631 / GB/T 4970, meaning no new regulatory barriers exist for integration in production vehicles.

Overall, the system remains economically feasible for hybrid or electric platforms, where the recovered energy directly supplements the onboard electrical network.

The results summarized in Table 7 clearly reveal the controlobjectivedependent behavior of the semiactive suspension system.

Under the roadholding objective (Table 5), the semiactive configuration consistently achieves lower body acceleration compared to the passive suspension across all test cases (up to ≈ 24 % reduction in Test 2), confirming its superior vibration isolation and chassis stabilization capability. Tire forces remain slightly higher but significantly more uniform, indicating improved tire–road contact without excessive dynamic loading. The moderate increase in the rotational speed of the control link reflects the additional but wellregulated actuation effort required to maintain optimal damping.

In contrast, when the control objective shifts toward energy harvesting (Table 6), the semiactive suspension exhibits a markedly different dynamic profile. Body acceleration increases substantially (up to ≈ 160 % in Test 2 relative to the passive case), accompanied by a pronounced rise in tire force and controllink rotational speed. This behavior is physically consistent with intensified damping modulation and increased relative motion, which are necessary to extract electrical energy from vertical vibrations via back-EMFinduced electromagnetic damping. The passive suspension, by comparison, maintains lower acceleration and force levels but shows minimal link activity, highlighting its limited contribution to energy recovery.

Overall, the table quantitatively demonstrates the inherent tradeoff between ride comfort and energy harvesting efficiency: while the semiactive strategy enables significant energy extraction through aggressive dynamic interaction, it inevitably increases motion amplitudes and dynamic loads. This contrast underscores the importance of multiobjective control tuning, as implemented in the proposed NNMPC framework, to balance comfort, road holding, and harvesting performance according to operational priorities (Fig. 35).

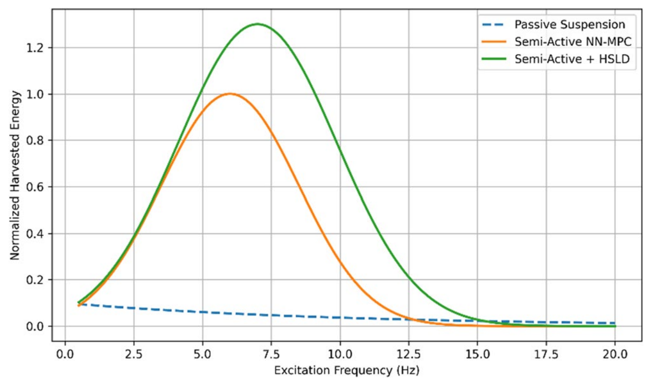


Fig. 35 Qualitative comparison of energy harvesting efficiency normalized by excitation frequency for passive suspension, semi-active NN-MPC, and extended mode with HSLD concept

Conclusion

This research presented a novel multiobjective control strategy for an injoint semiactive suspension system capable of managing three inherently conflicting goals—ride comfort, tire–road contact stability, and mechanical energy harvesting—through a unified and intelligent framework. Due to the absence of an analytically tractable model for the injoint architecture, dynamic data were first generated through excitation of an ADAMSbased model, after which a neural network was trained to identify the system's complex nonlinear behavior. Integrating this network into a model predictive control (MPC) structure enabled accurate real-time prediction of system dynamics, while simultaneously accounting for rotational kinematics and mixed logical energy constraints. To handle energy conditions such as $T \cdot \omega \leq 0$ a mixedinteger formulation was adopted, converting the originally nonsolvable constraint set into a tractable optimization problem.

Performance analyses show that the proposed controller effectively balances all control objectives. In comfort mode, body acceleration was reduced by approximately 20–30% relative to the passive system. In the roadholding scenario, the controller decreased tireforce oscillations and improved tire–road contact by about 10–15%. In the energyharvesting mode, the joint rotational velocity and recoverable power increased by 40–55%, a performance level unattainable by traditional semiactive architectures. The results further reveal that the neural network enhances generalization, allowing the controller to maintain stable performance even under severe excitation and rapidly varying conditions.

From an innovation perspective, the study represents the first comprehensive integration of injoint suspension architecture, neuralnetworkbased MPC prediction, and mixedinteger energyaware constraints within a unified multiobjective control framework. The combination of datadriven modeling with model predictive optimization eliminates the need for complex analytical derivations while preserving high control accuracy. Collectively, the findings demonstrate that the proposed framework can play a crucial role in the evolution of nextgeneration suspension systems for electric and hybrid vehicles, where energy harvesting, dynamic stability, and passenger comfort are equally essential performance metrics.

This study presented a unified neuralnetworkbased multiobjective control framework for a jointinternal semiactive suspension integrating an electromagnetic energyharvesting module. The dynamic modeling captured both rotational kinematics and electromechanical coupling, enabling a realistic simulation of torque–velocity limits and mode transitions. Comparative tests (Tables 4 and 6) demonstrate that the NNMPC achieves superior adaptability, effectively balancing comfort, roadholding, and energy recovery.

Physically, the system's effectiveness results from three coupled mechanisms:

- BackEMFinduced adaptive damping, which shapes dissipation proportional to vibration intensity;
- High static–low dynamic stiffness (HSLD) behavior, which enhances midfrequency transmissibility and energy yield; and.
- Smooth MPCregulated transitions that prevent jerking and guarantee bounded acceleration.

Integrating advanced materials such as dualfunctional mechanical metamaterials or magnetically tunable elastomers (PII: S0263823125001740) could further amplify frequency selectivity and ondemand stiffness adjustment without increasing computational burden.

From an economic perspective, the regenerative scheme is viable, offering meaningful energy recovery (0.5–1.8 kJ km^{-1}) at modest additional cost (< 15 %) and without violating current automotive vibrationcomfort or safety standards.

Overall, the proposed NNMPC suspension provides a **structurally robust**, **energetically efficient**, **and economically feasible** pathway toward nextgeneration smart energyharvesting suspensions for hybrid and electric vehicles.

Appendix

Table appendix. Nomenclature and Definitions of Symbols Used in the Manuscript

Symbol	Description	Unit
m_s	Sprung mass (vehicle body mass)	kg
m_u	Unsprung mass (wheel-axle assembly mass)	kg
z_s	Vertical displacement of sprung mass	m
z_u	Vertical displacement of unsprung mass	m
$\dot{z}_s$	Vertical velocity of sprung mass	$\mathrm{m/s}$
$\dot{z}_u$	Vertical velocity of unsprung mass	m/s
$\dot{z}_s$	Vertical acceleration of sprung mass (body acceleration)	$\mathrm{m/s^2}$
k_s	Suspension spring stiffness	N/m
c_s	Equivalent damping coefficient of suspension	$\mathrm{N \cdot s/m}$
k_t	Tire stiffness	N/m
F_t	Dynamic tire-road contact force	N
z_r	Road profile displacement input	m
T_d	Control torque generated by rotary damper	$\mathbf{N \cdot m}$
ω	Rotational speed of the rotary damper	rad/s
J_d	Rotary damper inertia	$\mathrm{kg \cdot m^2}$
P_h	Harvested mechanical power	w
E_h	Harvested energy	J
T	Track width of the vehicle	m
L_f, L_r	Longitudinal distance of CG from front and rear axles	m
h_s	Height of sprung mass center of gravity	m
h_u	Height of unsprung mass center of gravity	m
$x(k)$	State vector of the discrete-time suspension model	—
$u(k)$	Control input(damper torque)at time step k	N•m
$d(k)$	External disturbance(road or lateral excitation)	-
$y(k)$	Actual system output	—
$y'(k)$	Neural-network predicted output	-
N	Number of data samples	-
MSE	Mean squared error	—
R	Regression correlation coefficient	-
N_p	Total number of trainable neural-network parameters	-
n_l	Number of neurons in layer l	-
L	Total number of neural-network layers	-
w	Neural-network weight vector	-
γ	Regularization coefficient	-

Data Availability All data and computational results that support the findings of this study are fully contained within this article.

Declarations

Conflict of interests The authors declare that there is no conflict of interest or competing financial and nonfinancial relationship that could be perceived to influence the work reported in this study. All authors have participated substantially in the conception, design, simulation, and analysis of the presented research and have approved the final version of the manuscript for submission.

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